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RBMX2: A Pivotal Regulator Linking *Mycobacterium bovis* Infection to Epithelial-Mesenchymal Transition and Lung Cancer Progression

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eLife Assessment

The identification of RBMX2 as a novel regulator linking mycobacterial infection to Epithelial-Mesenchymal Transition and cancer progression are **fundamental** findings that advance our understanding of a major research question about the link between infectious and non-infectious diseases, microbiology and oncology. It does so by introducing RBMX2 as a novel host factor, a potential therapeutic target and biomarker for both TB and lung cancer. The evidence provided is **convincing** because it is appropriate and the validated multi-omics methodologies used are in line with the current state of the art. This study will be of interest to scientists working in the fields of drug discovery, microbiology and oncology.

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Abstract

Tuberculosis (TB) is a complex disease that arises from the dynamic interplay among the pathogen, host, and environmental factors. In 2022, TB affected approximately 10.6 million people worldwide, resulting in 1.3 million deaths. In regions with a high burden of zoonotic TB, Mycobacterium bovis (M. bovis) accounts for nearly 10% of human TB cases. The immune evasion strategies and latent nature of Mycobacterium tuberculosis (M. tb) pose significant challenges to understanding the host immune response to TB infection. In this study, we identify RNA-binding motif protein X-linked 2 (RBMX2) as a novel host factor that may facilitate *M. bovis* infection. However, the molecular mechanisms and functional roles of RBMX2 during M. bovis infection remain poorly understood.

Our findings show that RBMX2 is significantly upregulated across multiple cell types following infection, including embryonic bovine lung (EBL) cells, bovine macrophage (BoMac) cells, bovine pulmonary alveolar primary cells, and human alveolar epithelial cells (A549). Using a combination of global transcriptomic sequencing, proteomic profiling, cell adhesion assays, ChIP-PCR, and Western blotting, we demonstrate that RBMX2 suppresses cell adhesion and tight junction formation in EBL cells while promoting M. bovis adhesion and invasion through activation of the p65 signaling pathway. Importantly, integrated analyses of transcriptomic, proteomic, and metabolomic data reveal that RBMX2 plays a key role in regulating epithelial-mesenchymal transition (EMT), a cellular process closely associated with cancer progression.

To further explore the potential link between RBMX2 and cancer, we analyzed the TIMER2.0 database and found elevated RBMX2 expression in lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC) tissues compared to normal lung tissues. These findings were corroborated by immunofluorescence staining. Using an M. bovis-infected BoMac-induced EBL-EMT model, we show that RBMX2 promotes EMT through activation of the p65/MMP-9 pathway following infection. Additionally, we validated the EMT process in human lung epithelial cancer cells (H1299), further supporting the involvement of RBMX2 in this transition.

Collectively, our study identifies RBMX2 as a novel host factor that may enhance M. bovis infection and drive infection-induced EMT. These findings provide new insights into the molecular mechanisms underlying TB pathogenesis and suggest that RBMX2 could serve as a potential target for the development of TB vaccines and therapeutic strategies against TB-associated lung cancer.

Introduction

Tuberculosis (TB) remains a major global health threat, with approximately 10.6 million new cases and 1.3 million deaths reported worldwide in 2022 [1]. Mycobacterium tuberculosis (M. tb) and its zoonotic variant, Mycobacterium bovis (M. bovis), are responsible for a significant proportion of TB cases, particularly in regions with a high burden of zoonotic diseases, where M. bovis accounts for up to 10% of human TB infections. The immune evasion strategies of M. tb and its variants complicate our understanding of host immune responses and disease progression.

Recent studies have increasingly linked chronic infections, including TB, to cancer development [2], [3]. The persistent inflammatory environment induced by chronic infection can drive cellular transformations, such as epithelial-mesenchymal transition (EMT), which is closely associated with cancer initiation and metastasis [4]. However, the specific molecular mechanisms and host factors involved in this infection-cancer axis remain largely uncharacterized.

In this context, RNA-binding motif protein X-linked 2 (RBMX2) has emerged as a potential host factor involved in both TB infection and cancer progression. Our previous studies demonstrated that RBMX2 regulates alternative splicing of APAF-1 introns, thereby modulating apoptosis following M. bovis infection [5]. Furthermore, accumulating evidence suggests that its homolog, RBMX, plays dual and context-dependent roles in tumorigenesis, acting as either an oncogene or a tumor suppressor. For example, RBMX is overexpressed in hepatocellular carcinoma [6] and T-cell lymphomas [7], whereas its expression is reduced in pancreatic ductal adenocarcinoma, indicating a complex and tissue-specific regulatory function [8].

Despite these findings, the role of RBMX2 in M. bovis infection and its potential involvement in EMT regulation remain largely unexplored. Based on our preliminary data, we hypothesize that RBMX2 promotes M. bovis adhesion and invasion by modulating host cell properties, and further facilitates EMT through activation of the p65/MMP-9 signaling pathway. This study aims to elucidate the molecular mechanisms by which RBMX2 contributes to M. bovis infection and its potential role in infection-associated EMT, thereby providing novel insights into the intersection of TB and lung cancer. These findings may highlight RBMX2 as a promising therapeutic target for TB and TB-related malignancies.

Results

Elevated expression of RBMX2 in *Mycobacterium bovis*-infected cells and tuberculous Lesions

M. bovis was used to infect the EBL cell library, which was constructed using CRISPR/Cas9 technology in our laboratory, at a multiplicity of infection (MOI) of 100:1. We evaluated cell survival and performed high-throughput sequencing on the viable cells to identify key host factors potentially involved in anti-M. bovis defense. Notably, RBMX2 knockout cells exhibited significant resistance to infection.

Further analysis using AlphaFold Multimer revealed that amino acid residues 56 – 134 of RBMX2 contain an RNA recognition motif (**Supplementary Fig. 1A**). Additionally, we identified 611 RBM family sequences at the bovine protein exon level and classified them into subfamilies based on motif similarity. RBMX2 was predominantly assigned to subfamily VII, which includes proteins such as EIF3G, RBM14, RBM42, RBMX44, RBM17, PUF60, SART3, and RBM25 (**Supplementary Fig. 1B**). Comparative analysis of amino acid sequences indicated a high degree of conservation of RBMX2 across species (**Supplementary Table 1**).

To further validate RBMX2 expression in EBL cells following infection with M. bovis and M. bovis Bacillus Calmette – Guerin (BCG), RNA samples were collected at 24, 48, and 72 hours post-infection (hpi). Real-time quantitative polymerase chain reaction (RT-qPCR) results showed that RBMX2 expression was significantly upregulated after infection with both M. bovis and M. bovis BCG (**Figs. 1A** and **B**). Moreover, increased RBMX2 expression was observed in bovine monocyte-derived macrophages (BoMac cells), bovine alveolar type II primary cells, and human lung epithelial cells (A549 cells) following M. bovis infection (**Figs. 1C–E**). Furthermore, we used WB to verify the expression levels of RBMX2 protein in EBL cells after M. bovis infection. The results showed that M. bovis infection upregulates the expression of RBMX2 protein in EBL cells. (**Fig. 1F**).

RBMX2 did not affect cell cycle but inhibited epithelial cell survival during *M. bovis* infection

In generating *RBMX2* monoclonal knockout (KO) cells, monoclonal cells were selected through limited dilution in 96-well plates. Sanger sequencing was conducted to identify different monoclonal cells with distinct knockout sites (**Supplementary Fig. 2A**). We validated the effect of RBMX2 knockout on the cell cycle using flow cytometry, and the results indicated that RBMX2 does not affect the cell cycle.(**Supplementary Fig. 2B**).

To investigate the potential role of *RBMX2* in resistance to *M. bovis* infection, we assessed the survival rate of EBL cells and H1299 cells at different hours post-infection using CCK-8 assay. The findings revealed an enhanced survival rate in both *RBMX2* knockout monoclonal EBL cells following *M. bovis* infection (**Fig. 1G**). Furthermore, RBMX2-silenced H1299 cells exhibited a

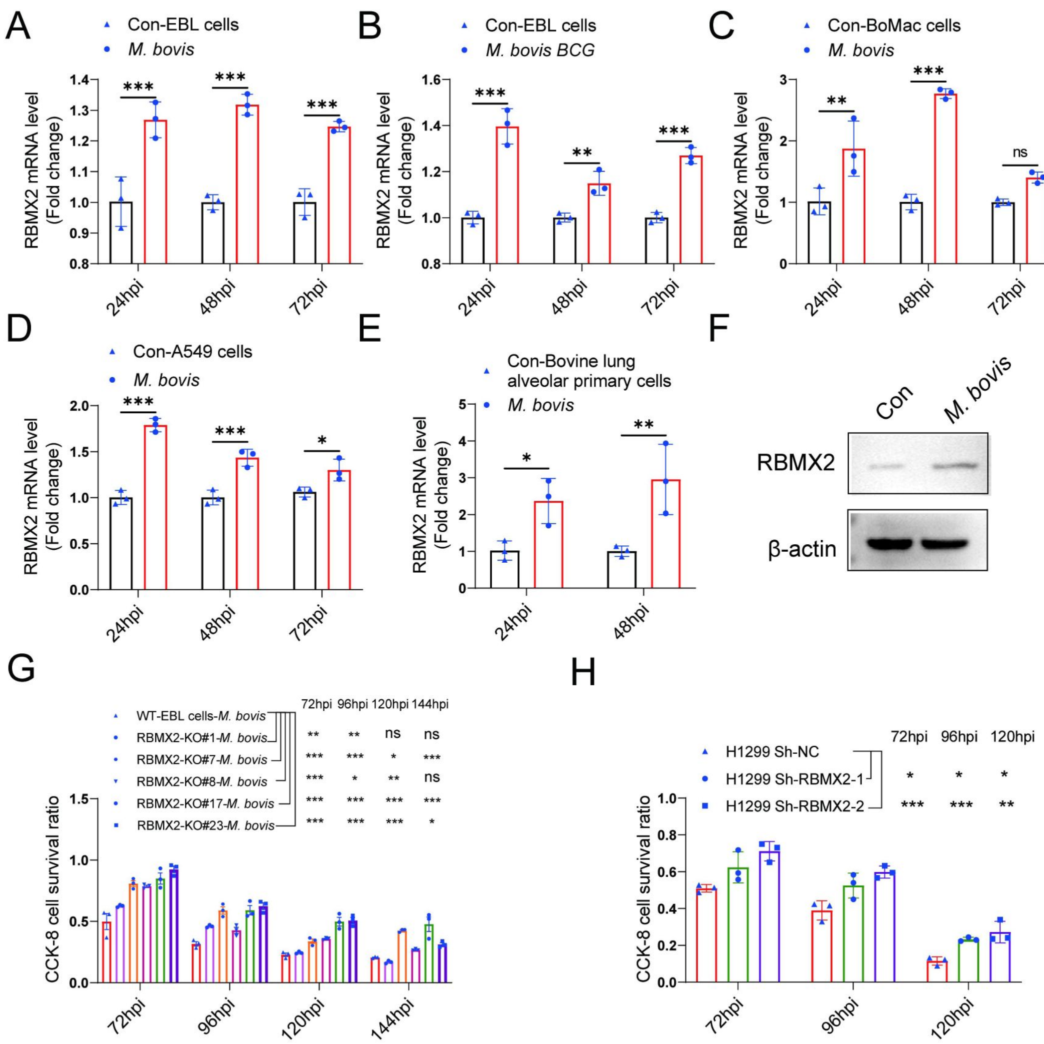


Figure 1.

Expression of *RBMX2* after infection, and *RBMX2* did not affect cell proliferation but inhibited cell survival during *M. bovis* infection.

(A, B) The expression of *RBMX2* in EBL cells infected by (A) *M. bovis* and (B) *M. bovis* BCG were analyzed by RT-qPCR. Data were represented by fold expression relative to uninfected cells.

(C,-E) The expression of *RBMX2* mRNA in (E) BoMac cells, (F) A549 cells, and (G) bovine lung alveolar primary cells infected by *M. bovis* were analyzed via RT-qPCR. Data were represented by fold expression relative to uninfected cells.

(F) The expression of *RBMX2* in EBL cells infected by *M. bovis was* analyzed by WB. Data were represented by fold expression relative to uninfected cells.

(G) Detection of the ability of different *RBMX2* knockout site Monoclonal EBL cells against *M. bovis* infection by CCK-8 assay. Data were represented by the absorbance value relative to WT EBL cells after *M. bovis* infection.

(H) Detection of the ability of *RBMX2* slicing H1299 cells against *M. bovis* infection by CCK-8 assay. Data were represented by the absorbance value relative to Sh-NC H1299 cells after *M. bovis* infection

One-way ANOVA and two-way ANOVA were used to determine the statistical significance of differences between different groups. Ns presents no significance, * presents $p < 0.05$, **presents $p < 0.01$, and *** presents $p < 0.001$ indicate statistically significant differences. Data were representative of at least three independent experiments.

higher survival rate compared to H1299 ShNc cells after *M. bovis* infection (**Fig. 1H**). Notably, *RBMX2* knockout in EBL cells and silencing in H1299 cells demonstrated significant improvements in cells survival at 96 and 120 hpi compared to Wild Type (WT) EBL cells and H1299 ShNC cells, as evidenced by crystal violet staining (**Supplementary Fig. 2C**).

RBMX2-regulated genes associated with cell tight junction and EMT-related pathways

To investigate the role of RBMX2 knockout during M. bovis infection, we selected RBMX2 knockout EBL cells and wild-type (WT) EBL cells at 0 hours post-infection (hpi) (2 hpi post-gentamicin, referred to as 0 hpi), 24 hpi, and 48 hpi for RNA-Seq analysis. A total of 16,079 genes were identified with a significance threshold of $p < 0.05$ and a log2(fold change) > 2 relative to WT EBL cells. Notably, 42 genes were significantly regulated at all three time points (0, 24, and 48 hpi) (**Fig. 2A**). Of these, 11 genes were significantly upregulated, while 31 were significantly downregulated. A heat map illustrating the expression levels of these genes across all samples is presented, with each row representing a gene and each column representing a sample (**Fig. 2A**).

To further explore the functional implications of these differentially expressed genes (DEGs) in RBMX2 knockout EBL cells during infection, Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analyses were conducted. The most significantly enriched biological processes included epithelial cell differentiation, cell adhesion, and biological adhesion. The most affected cellular components were the extracellular region, extracellular region part, extracellular space, and plasma membrane part. Molecular functions were primarily associated with activin receptor activity, type I (**Fig. 2B**). KEGG pathway analysis revealed that these DEGs were involved in several pathways, including ECM – receptor interaction, cGMP-PKG signaling pathway, PI3K-Akt signaling pathway, cytokine-cytokine receptor interaction, and vascular smooth muscle contraction (**Fig. 2C**).

Proteomic analysis of 48 hpi protein samples further validated the phenotypes related to cell tight junctions and epithelial-mesenchymal transition (EMT). GO analysis identified significant enrichment in pathways related to extracellular space, integrin binding, basement membrane, and glucose homeostasis, whereas KEGG analysis highlighted pathways such as cell adhesion molecules, ECM–receptor interaction, and the TGF-beta signaling pathway (**Figs. 2D** and **2E**).

We also assessed how varying infection durations influenced RBMX2 knockout EBL cells using GO analysis. At 0 hpi, differentially expressed genes were primarily associated with cell junctions, extracellular regions, and cell junction organization (**Supplementary Fig. 3A**). At 24 hpi, enriched pathways included basement membrane, cell adhesion, integrin binding, and cell migration (**Supplementary Fig. 3B**). By 48 hpi, genes were mainly annotated to epithelial cell differentiation and were negatively regulated during epithelial cell proliferation (**Supplementary Fig. 3C**), suggesting that RBMX2 modulates cellular connectivity throughout the stages of M. bovis infection.

KEGG analysis revealed distinct enrichment patterns over time. At 0 hpi, genes were enriched in the MAPK signaling pathway, chemical carcinogen-DNA adducts, and chemical carcinogen–receptor activation (**Supplementary Fig. 3D**). At 24 hpi, significant enrichment was observed in ECM – receptor interaction, PI3K – Akt signaling pathway, and focal adhesion (**Supplementary Fig. 3E**). At 48 hpi, enrichment was noted in the TGF-beta signaling pathway, transcriptional misregulation in cancer, microRNAs in cancer, small cell lung cancer, and the p53 signaling pathway (**Supplementary Fig. 3F**).

To validate the RNA-seq results, we selected 10 genes at random for RT-qPCR analysis (**Fig. 2F** and **Supplementary Fig. 4C**). The mRNA levels of these genes were consistent with RNA-seq data. These genes included FRMD4B, MCTP1, HSPB8, ST14, and OMD, which are associated with

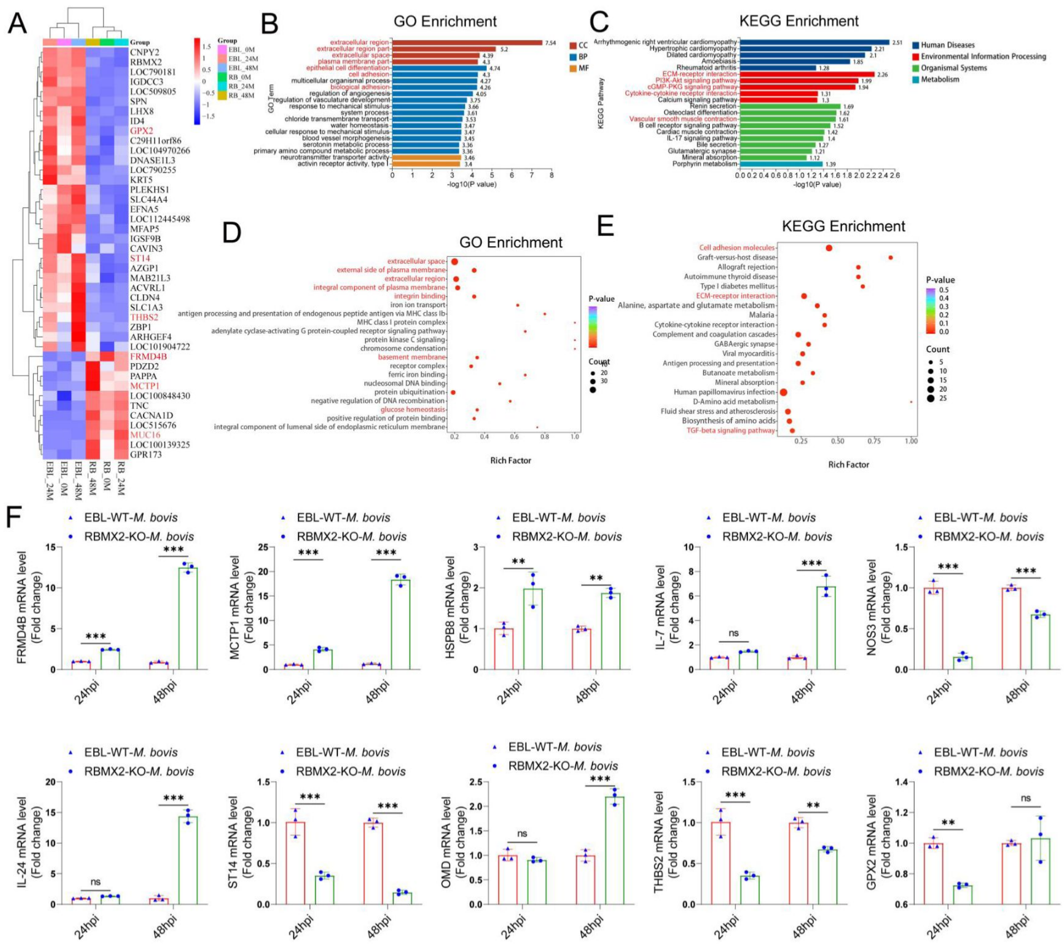


Figure 2.

Transcriptome and proteomic analysis in *RBMX2* knockout and WT EBL cells after *M. bovis* infection.

(A) The heat map illustrates some genes that had been all enriched in RBMX2 knockout and WT EBL cells after *M. bovis* infection in 0 (2 hpi post-gentamicin, recorded as 0 hpi), 24, and 48hpi. Red represents upregulated genes, and blue represents downregulated genes. Each group represented three independent samples.
(B) GO Analysis of all enriched genes in 0 (2 hpi post-gentamicin, recorded as 0 hpi), 24 and 48hpi. Data were represented as all enriched pathways in RBMX2 knockout EBL cells relative to WT EBL cells after *M. bovis* infection thrice.
(C) KEGG Analysis of all enriched genes in 0 (2 hpi post-gentamicin, recorded as 0 hpi), 24, and 48hpi. Data were represented as all enriched pathways in RBMX2 knockout EBL cells relative to WT EBL cells after *M. bovis* infection in three times.
(D) GO Analysis of all enriched proteins in 48hpi. Data were represented as all enriched pathways in RBMX2 knockout EBL cells relative to WT EBL cells after *M. bovis* infection.
(E) KEGG Analysis of all enriched genes in 48hpi. Data were represented as all enriched pathways in RBMX2 knockout EBL cells relative to WT EBL cells after *M. bovis*-infection (MOI 20) in three times.
(F) Identification the expression of related genes mRNA enriched by RT-qPCR. Data were represented as the fold expression in RBMX2 knockout EBL cells relative to WT EBL cells after *M. bovis* infection.
Two-way ANOVA was used to determine the statistical significance of differences between different groups. Ns presents no significance, * presents $p < 0.05$, **presents $p < 0.01$, and *** presents $p < 0.001$ indicate statistically significant differences.

scaffold proteins, cell adhesion, and epithelial barrier function. Additionally, MCTP1, HSPB8, IL-24, IL-7, GPX2, and NOS3 were linked to apoptosis, inflammation, and oxidative stress.

RBMX2 interferes with the integrity of tight junctions in epithelial cells caused by *M. bovis*

Through transcriptome sequencing, we identified 41 genes that underwent transcript changes following *M. bovis* infection. Of these, 22 genes were upregulated, while 19 genes were downregulated (**Supplementary Figs. 4A** and **B**). The downregulated genes were primarily associated with tight junction and leukocyte transendothelial migration (**Fig. 3A**). In contrast, the upregulated genes were mainly linked to immunity-related pathways, including the defense response to viruses and regulation of viral processes (**Fig. 3B**).

Based on these finding,we hypothesized that *RBMX2* could damage the integrity of epithelial cell tight junction. To validate this prediction, we constructed a model of *M. bovis*-destroyed tight junctions in EBL cells. EBL cells were infected with *M. bovis* and we subsequently performed RT-qPCR and Western blot (WB) analyses to assess the expression of cell tight junction-related mRNAs (*TJP1*, *OCLN*, *CLDN-5*, and *CLDN-7*) and proteins (ZO-1, OCLN, and CLDN-5). We found that the expression levels of most mRNAs and proteins were significantly reduced following *M. bovis* infection compared to the uninfected coutrol (**Figs. 3C** and **D**).

Tight junctions are critical intercellular adhesion structures that define the permeability of barrier-forming epithelial cells [9]. The ability of epithelial cells to maintain functional intercellular adhesion is essential for forming a tight epithelial protective barrier [10]. To further investigate the effect of *M. bovis* on epithelial cell adhesion, we conducted a cell adhesion assay, revealing a significant decrease in the intercellular adhesion ratio (**Fig. 3E**). These results validated that *M. bovis* can compromise the tight junction of EBL cells.

To explore the impact of *RBMX2* on the epithelial cells barrier, we infected RBMX2 knockout and WT EBL cells with *M. bovis* and assessed the expression of epithelial barrier-related mRNAs (*TJP1*, *OCLN*, and *CLDN-5*) and proteins (ZO-1, OCLN, and CLDN-5) using RT-qPCR and WB. The *RBMX2* knockout EBL cells exhibited a significant upregulation of all three proteins compared to WT EBL cells (**Figs. 3F** and **G**). The cell adhesion assay also demonstrated that *RBMX2* knockout EBL cells displayed a heightened adhesion ratio relative to WT EBL cells (**Fig. 3H**).

Tight junctions are essential component of the epithelial barrier and can be compromised by bacteria infections, contributing to inflammation [11], [12]. To investigate whether *RBMX2* mediates intracellular inflammation by regulating the tightness of the epithelial barrier, we measured the mRNA levels of inflammatory factors (IL-1β, TNF, and IL-6) in *RBMX2* knockout EBL cells post-infection. The results showed a decrease in the expression of these inflammatory factors in RBMX2 knockout EBL cells compared to WT EBL cells after *M. bovis* infection (**Fig. 3I**).

In summary, our findings indicate that *RBMX2* affects the integrity of epithelial cell tight junctions, and knocking out *RBMX2* enhances tight junction formation while decreasing the inflammatory response induced by *M. bovis*.

RBMX2 disrupted the epithelial barrier via activating *p65*

The MAPK pathway and NF-kappaB pathway play crucial roles in regulating tight junctions between epithelial cells [13]–[16]. To further elucidate the regulatory mechanism of *RBMX2* in the epithelial barrier's tight junctions, we found that the MAPK pathway was enriched in our transcriptome sequencing data, and the potential regulatory transcription factor p65 was predicted through the JASPAR database (**Supplementary Table 2**).

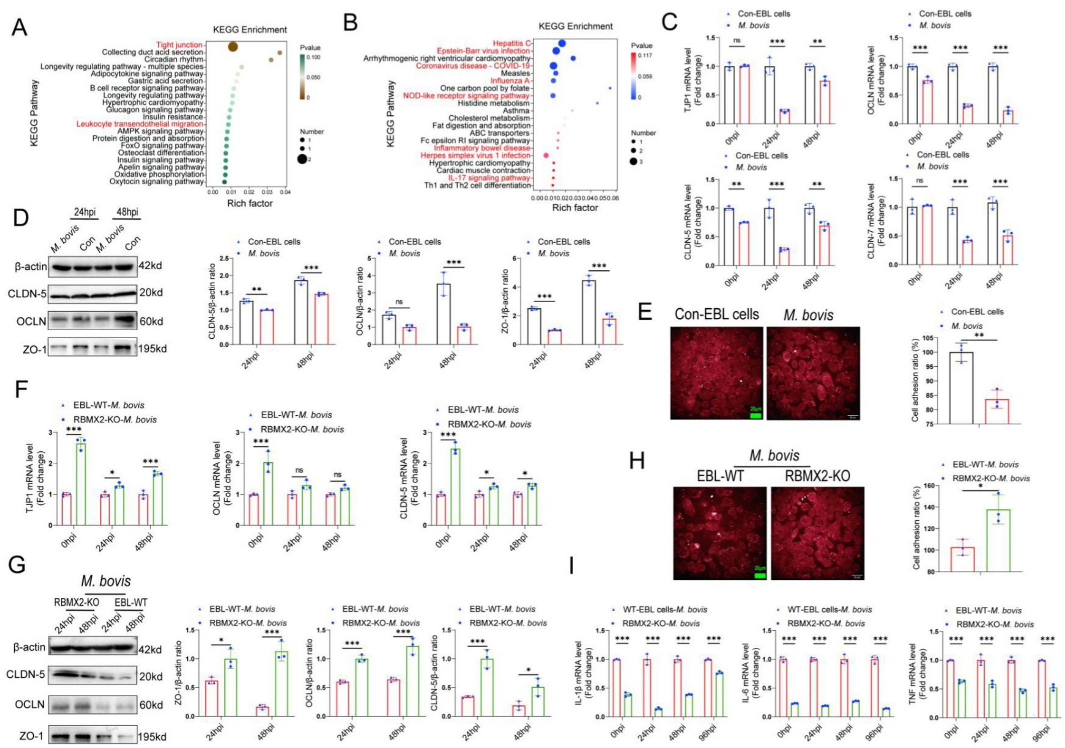


Figure 3.

RBMX2 had the potential to induce the disruption of tight junctions in EBL cells after *M. bovis* infection.

(A, B) KEGG analysis was conducted to identify the (A) down-regulated and (B) up-regulated pathways among the enriched genes after *M. bovis* infection of WT EBL cells. Data were relative to WT EBL cells without *M. bovis* infection
(C, D) The expression of epithelial cells tight junction-related (C) mRNAs (*TJP1*, *CLDN-5*, *CLDN-7*, and *OCLN*) and (D) proteins (ZO-1, CLDN-5, and OCLN) were assessed after *M. bovis* infection of WT EBL cells via RT-qPCR and WB. Data were relative to WT EBL cells without *M. bovis* infection.
(E) Cell adhesion ratio was evaluated via cell adhesion assay after WT EBL cells were infected with *M. bovis* using High-content imaging. Data were relative to WT EBL cells without *M. bovis* infection. Scale bar: 20 μm.
(F, G) The expression of epithelial tight junction-related (F) mRNAs (*TJP1*, *CLDN-5*, and *OCLN*) and (G) proteins (ZO-1, CLDN-5, and OCLN) in *RBMX2* knockout EBL cells after *M. bovis*-infection through RT-qPCR and WB. Data were relative to WT EBL cells with *M. bovis* infection.
(H) Cell adhesion assay was conducted to assess the cell adhesion ratio of *RBMX2* knockout EBL cells after infection with *M. bovis*. Data were relative to WT EBL cells with *M. bovis* infection. Scale bar: 20 μm.
(I) Expression of inflammatory factors-related factors (*IL-6*, *IL-1β*, and *TNF*) were assessed after *RBMX2* knockout EBL cells infected by *M. bovis*. Data were relative to WT EBL cells with *M. bovis* infection
T-test and Two-way ANOVA were used to determine the statistical significance of differences between different groups. Ns presents no significance, * presents $p < 0.05$, **presents $p < 0.01$, and *** presents $p < 0.001$ indicate statistically significant differences. Data were representative of at least three independent experiments.

We assessed the phosphorylation levels of MAPK pathway proteins and p65 in RBMX2 knockout EBL cells, revealing a significant reduction in the phosphorylation of p65 and MAPK/p38/JNK in these cells following M. bovis infection compared to wild-type (WT) EBL cells (**Figs. 4A** and **B**).

To determine whether RBMX2 specifically regulates MAPK/p38/JNK proteins and p65 to promote disruption of tight junctions and reduce intercellular adhesion, RBMX2 knockout EBL cells were treated with PMA (a p65 activator), Anisomycin (a JNK activator), and ML141 (a p38 activator) for 12 h prior to M. bovis infection. The optimal concentrations of these activators were determined using Western blotting (WB) and CCK-8 cell viability assays, resulting in concentrations of 100 nM for PMA, 10 μ M for Anisomycin, and 10 μ M for ML141, with no significant effects on cell viability (**Supplementary Figs. 5A, B**, and **C**). WB results indicated that activation of p65 and the MAPK/p38 pathways suppressed the expression of tight junction proteins, including ZO-1, OCLN, and CLDN-5, following M. bovis infection (**Figs. 4C** and **D**).

To investigate whether RBMX2 regulates the cell adhesion ratio via p65 and MAPK/p38/JNK pathways, we examined the effects of p65, p38, and JNK activators on cell adhesion in RBMX2 knockout EBL cells. Cell adhesion assays demonstrated that treatment with the p65 activator significantly reduced the adhesion ratio compared to treatments with the p38 and JNK activators (**Figs. 4E** and **F**).

To further confirm the association between p65 and tight junction regulation in WT EBL cells after M. bovis infection, we used siRNA to knock down p65 expression. WB analysis showed that siRNA effectively reduced both p65 and phosphorylated p65 (p-p65) expression (**Figs. 4G and H**). Furthermore, WB results revealed that p65 suppression enhanced the expression of ZO-1, OCLN, and CLDN-5 following M. bovis infection (**Figs. 4G** and **H**). Additionally, silencing p65 in WT EBL cells inhibited M. bovis invasion into epithelial cells, as demonstrated by plate counting assays (**Fig. 4I**).

Finally, to investigate the relationship between *RBMX2*-mediated p65 and the findings above, EBL cells were infected with *M. bovis* and *M. bovis* BCG. WB and high-content live imaging system outcomes demonstrated diminished nuclear translocation of the p65 protein in *RBMX2* knockout EBL cells compared with WT EBL cells (**Figs. 4J** and **K**).

In summary, our findings validate that *RBMX2* can regulate the phosphorylation and nuclear translocation of p65 protein, leading to the degradation of tight junction proteins in EBL cells infected with *M. bovis*.

RBMX2 promotes adhesion, invasion, and intracellular survival of pathogens

Disrupting the intercellular tight junction barrier enhances bacterial adhesion and invasion [17,18]. Based on these findings, we hypothesize that *RBMX2* may facilitate the adhesion and invasion of *M. bovis* by degrading the tight junction proteins in EBL cells. To test this hypothesis, EBL cells were infected with *M. bovis* at an MOI of 20:1, plate count assays indicated a significant reduction in *M. bovis* adhesion in *RBMX2*-KO#17 and KO#23 EBL cells compared with WT EBL cells at 15 min, 30 min, 1 h, and 2 h post-infection (**Supplementary Fig. 6A**). Furthermore, the invasion of *M. bovis* in *RBMX2*-KO#17 and KO#23 EBL cells were decreased at 2, 4, and 6 h post-infection compared with WT EBL cells (**Supplementary Fig. 6B**). Additionally, the intracellular survival of *M. bovis* in *RBMX2*-KO#17 and KO#23 EBL cells was lower at 24, 48, 72 and 96 hpi compared with WT EBL cells (**Supplementary Fig. 6C**). Silencing *RBMX2* in human lung epithelial cells (H1299) also led to a significant reduction in the adhesion, invasion, and intracellular survival of *M. bovis* (**Supplementary Figs. 6D-F**). Thus, the host factor *RBMX2* is critical for promoting the adhesion, invasion, and intracellular survival of *M. bovis*. In addtion,

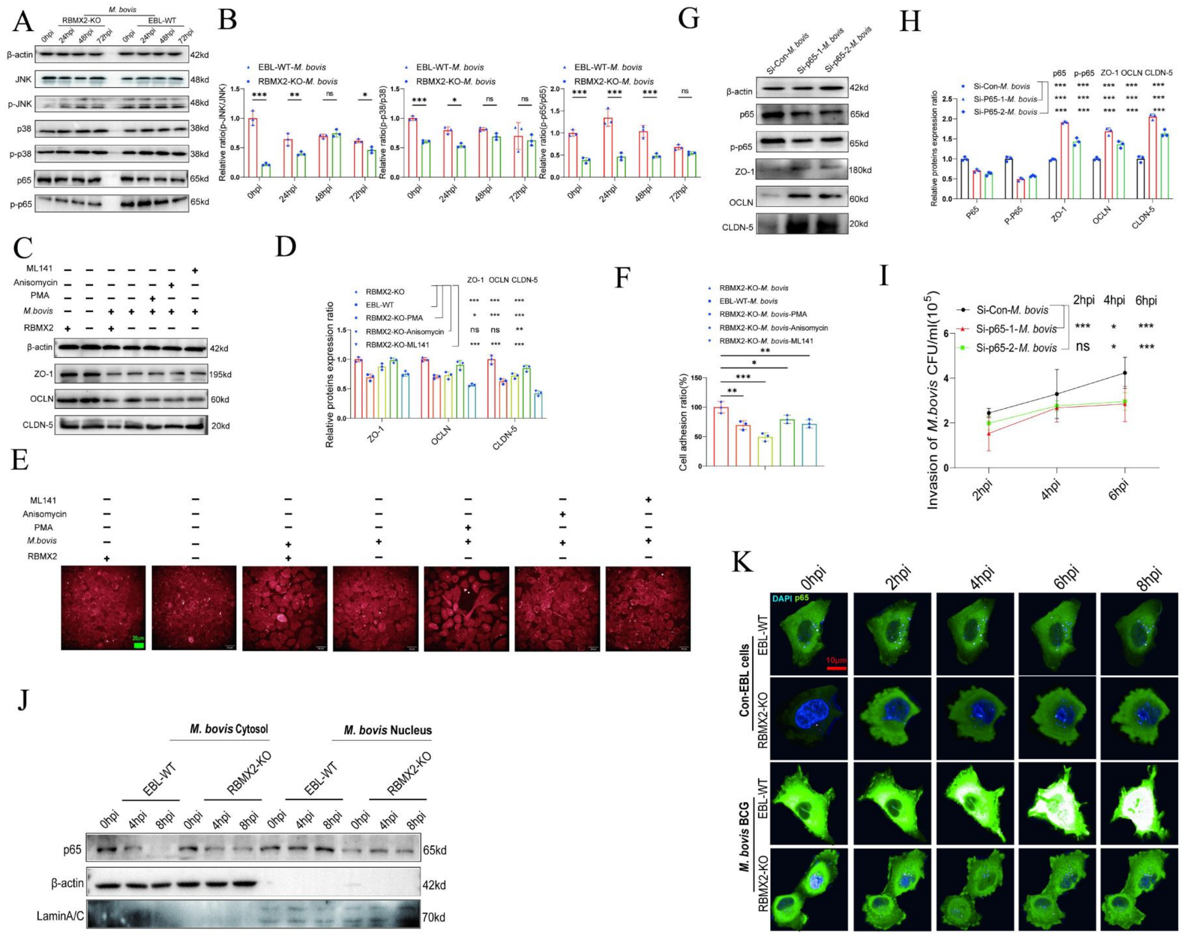


Figure 4.

***RBMX2* facilitated the disruption of epithelial tight junctions through the promotion of p65 protein phosphorylation and translocation and then enhanced the processes of *M. bovis* adhesion, invasion, and intracellular survival.**

(A, B) Activation of the MAPK pathway-related protein and p65 protein were activated after *RBMX2* knockout and WT EBL cells infected by *M. bovis* via WB. Data were relative to WT EBL cells with *M. bovis* infection.

(C, D) Expression of tight junction-related proteins (ZO-1, CLDN-5, and OCLN) was assessed in *RBMX2* knockout EBL cells treated with three p38/p65/JNK pathways activators after *M. bovis* infection via WB. Data were relative to *RBMX2* knockout EBL cells untreated activators with *M. bovis* infection.

(E, F) Evaluate the impact of three *p38/p65/JNK* pathways activators on the ratio of intercellular adhesion via cell adhesion assay. Data were relative to *RBMX2* knockout EBL cells untreated activators with *M. bovis* infection. Scale bar: 20 μm.

(G, H) Evaluate the silencing efficiency of siRNA on p65 protein expression and its impact on the expression of ZO-1, CLDN-5, and OCLN proteins through WB. Data were relative to SiRNA-NC in WT EBL cells with *M. bovis* infection.

(I) The effect of p65 silencing on the invasive ability of *M. bovis* in WT EBL cells. Data were relative to SiRNA-NC in WT EBL cells with *M. bovis* infection.

(J) The effect of *RBMX2* on the nuclear translocation of p65 protein after *M. bovis* infection using WB. β-actin presents cytosol and Lamin A/C presents nucleus. Data were relative to RBMX2 knockout EBL cells after *M. bovis* infection.

(K) The effect of *RBMX2* on the nuclear translocation of p65 protein after *BCG*-infection using High-content real-time imaging. Using pCMV-EGFP-p65 plasmid transfect RBMX2 knockout and WT EBL cells. The nucleus is stained with blue fluorescence. Data were relative to WT EBL cells without BCG infection.

One-way ANOVA and two-way ANOVA were used to determine the statistical significance of differences between different groups. Ns presents no significance, * presents $p < 0.05$, **presents $p < 0.01$, and *** presents $p < 0.001$ indicate statistically significant differences. Data were representative of at least three independent experiments.

pathway activators were employed to investigate the relationship between pathway activation and the regulation of *M. bovis* adhesion and invasion by *RBMX2*. The results showed that applying a p65 activator to *RBMX2* knockout EBL cells significantly impaired the tight junction function of the epithelial barrier, thereby enhancing *M. bovis* adhesion and invasion in EBL cells by regulating the phosphorylation and nucleation of p65 (**Supplementary Figs. 7A** and **B**). Hence, RBMX2 promotes *M. bovis* adhesion and invasion in EBL cells by regulating the phosphorylation and nucleation of p65.

To further investigate the role of RBMX2 in adhesion and invasion across different virulent mycobacterial species, we employed both attenuated virulent Mycobacterium bovis BCG and Mycobacterium smegmatis (M. smegmatis) to infect EBL cells. Plate count assays demonstrated a significant decrease in the adhesion of both M. bovis BCG and M. smegmatis in RBMX2 knockout EBL cells at multiple time points post-infection compared to wild-type (WT) EBL cells (**Supplementary Figs. 7C** and **D**). Moreover, the invasion of both mycobacterial species was notably reduced at various hours post-infection in RBMX2 knockout cells (**Supplementary Figs. 7E** and **F**). These findings suggest that the virulence of Mycobacterium does not influence the ability of RBMX2 to facilitate adhesion and invasion in EBL cells.

Additionally, to assess whether RBMX2 regulates interactions with other pathogens associated with bovine pneumonia, EBL cells were infected with Salmonella and Escherichia coli (E. coli). Our results revealed that RBMX2 knockout EBL cells exhibited a reduction in bacterial adhesion and invasion across all tested species compared to WT cells (**Supplementary Figs. 7G–J**).

Collectively, these experiments provide strong evidence that RBMX2 broadly enhances the adhesion and invasion of pathogens linked to bovine pneumonia, underscoring its potential as a therapeutic target for developing disease-resistant cattle.

RBMX2 is highly expressed in lung cancer and regulates cancer-related metabolites

Previous studies have demonstrated a link between bacterial infection and the initiation of EMT [19–21]. Prolonged infection with *M. tb* or *M. bovis* induces oxidative stress, activates Toll-like receptors (TLR), and elicits the release of inflammatory cytokines [22–24]. Consequently, this phenomenon creates a favorable microenvironment for tumor development, progression, and dissemination [25, 26]. Epidemiological investigations have established a correlation between TB and the occurrence of lung cancer [27–30]. However, the precise cellular mechanism underlying this association remain unclear.

In our research, we conducted a comprehensive analysis of gene families, revealing a remarkable degree of *RBMX2* conservation among bovine, monkey, and human sources (**Fig. 5A**). To further elucidate the involvement of *RBMX2* in cancer, we utilized the *TIMER2.0* database to assess its expression patterns across various cancer types within The Cancer Genome Atlas (TCGA). Our findings indicated a notable upregulation of *RBMX2* in lung cancer, specifically in lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC) (**Fig. 5B**).

To investigate the elevated expression of *RBMX2* in lung cancer further, we measured its expression in both lung cancer epithelial cell lines and normal lung epithelial cell lines using RT-qPCR. The results demonstrated that *RBMX2* expression in lung cancer epithelial cell lines (NCIH1299, NCIH460, and CALU1) was significantly higher than that in normal lung epithelial cells (BEAS-2B) (**Fig. 5C**). Additionally, immunofluorescence (IF) analysis revealed that *RBMX2* expression levels were also higher in LUAD and LUSC tumor tissues compared to adjacent non-cancerous lung tissues (**Figs. 5D-F**).

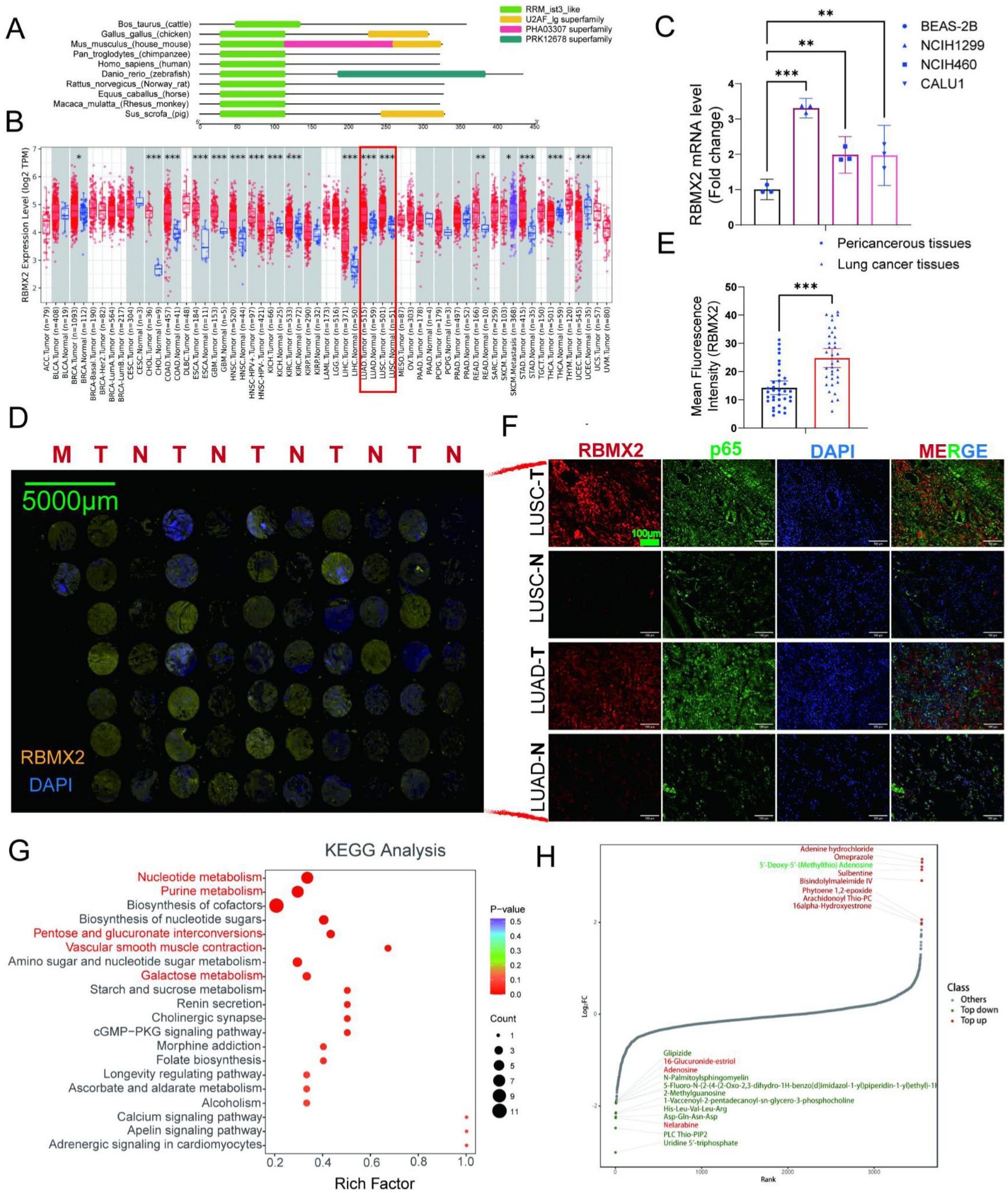


Figure 5.

RBMX2 is highly expressed in tumor tissues and regulates cancer-related metabolites

(A) A comparative analysis of the functional domains of the *RBMX2* protein across ten different species.
(B) Analyzing the expression patterns of *RBMX2* in pan cancer using *TIMER2.0* cancer database.
(C) The expression of *RBMX2* in different lung cancer cells and normal lung epithelialcells via RT-qPCR. Data were relative to normal lung epithelialcells (BEAS-2B)
(D-E) The expression of *RBMX2* in lung cancer clinical tissues via IF. RBMX2 is stained with yellow fluorescence, and the nucleus is stained with blue fluorescence. Data were relative to pericancerous lung tissues. Scale Bar: 5000 μm.
(F) The expression of *RBMX2* and p65 in LUAD and LUSC clinical tissues via IF. *RBMX2* is stained with red fluorescence, p65 is stained with green fluorescence, and the nucleus is stained with blue fluorescence. Data were relative to normal lung tissues. Scale Bar: 100 μm.
(G) KEGG analysis of differential metabolite enrichment pathways in *RBMX2* knockout EBL cells compared to WT EBL cells after *M. bovis* infection.
(H) Dynamic distribution map of top 20 differential metabolites in RBMX2 knockout EBL cells compared to WT EBL cells after *M. bovis* infection.

Furthermore, we investigate whether *RBMX2* regulates specific EMT-related metabolites to mediate the EMT process in EBL cells following *M. bovis* infection. Our metabolomic analysis of EBL cell samples at 48 hpi revealed that *RBMX2* knockout primarily enriched pathways related to nucleotide metabolism, biosynthesis of cofactors, biosynthesis of nucleotide sugars, pentose and glucuronate interconversions, vascular smooth muscle contraction, amino sugar, and nucleotide sugar metabolism, chemokine signaling pathway, aldosterone synthesis and secretion, and cGMP-PKG signaling pathway. Thes pathways are associated with tumor cells proliferation, migration, and invasion (**Fig. 5G**). Differential metabolites are related to tumor metabolism, mainly including 6-glucuronic acid estriol, adenosine, uridine 5 ’- triphosphate, and 5’ - deoxy-5 ’-(methylthio) (**Figure 5H**).

RBMX2 promotes the transformation of EBL cells from epithelial cells to mesenchymal cells

To investigate the association between *RBMX2* and EMT induced by *M. bovis*, we intially infected EBL cells directly with *M. bovis*. This infection did not effectively induce the expression of mesenchymal cell marker proteins (**Supplementary Fig. 8A**). Previous literature indicates that *M. bovis* infection in macrophages, along with its secreted proteins, can stimulate the production of cytokines such as TNF, IL-6, and TGF-β, all of which promote EMT process [31–34].

To further explore the impact of *M. bovis* infection on EMT progression in EBL cells, we constructed a coculture model using *M. bovis*-infected BoMac cells (**Fig. 6A** and **Supplementary Fig. 8B**). In this model, we identified EMT-related cytokines, including IL-6 and TNF, in *M. bovis*-infected BoMac cells, revealing a significant increase compared to uninfected BoMac and EBL cells (**Fig. 6B**). Notably, *RBMX2* expression was upregulated in EBL cells within this coculture model following *M. bovis* infection (**Fig. 6C**).

We then examined the morphology changes in epithelial cells within the coculture model using electron microscopy. The results demonstrated that most cells transitioned from circular to an elongated morphology (**Fig. 6D**). Staining of the EBL cells cytoskeleton with ghost pen cyclic peptide highlighted significant morphological alterations, with EBL cells transforming from spherical to spindle-shaped (**Supplementary Fig. 8C**).

Additionally, we assessed the expression of EMT-related mRNAs and proteins in EBL cells after coculture with *M. bovis-infected* BoMac cells. This analysis reveald a significant downregulation of the epithelial cell marker protein *E-cadherin* and a notable upregulation of mesenchymal markers *N-cadherin* and *MMP-9*, as determined by RT-qPCR and WB (**Fig. 6E**, **Supplementary Fig. 8D**). The coculture with, *M. bovis-infected* BoMac cells enhanced the migratory and invasive properties of EBL cells, as demonstrated by Transwell assay (**Figs. 6F** and **G**).

To confirm the impact of *RBMX2* knockout on the EMT process in the coculture model, we observed that *RBMX2* knockout cells exhibited significant upregulation of the epithelial cell marker protein *E-cadherin* and downregulation of the mesenchymal cell marker *N-cadherin* and *MMP-9* compared to WT EBL cells, as shown by RT-qPCR and WB (**Fig. 6H** and **Supplementary Fig. 7E**). Wound healing and Transwell assays demonstrated that the migration and invasion rates of *RBMX2* knockout cells were markedly lower than those of WT EBL cells (**Figs. 6I, J**, and **K**).

Finally, we silenced RBMX2 in the human lung cancer epithelial cell line H1299, which expresses high levels of RBMX2, to assess the effect on EMT-related proteins, as well as invasion and migration ability following *M. bovis* infection. The results indicated that RBMX2 significantly inhibited the EMT process in H1299 cells post-infection (**Supplementary Figs. 9A** and **B**).

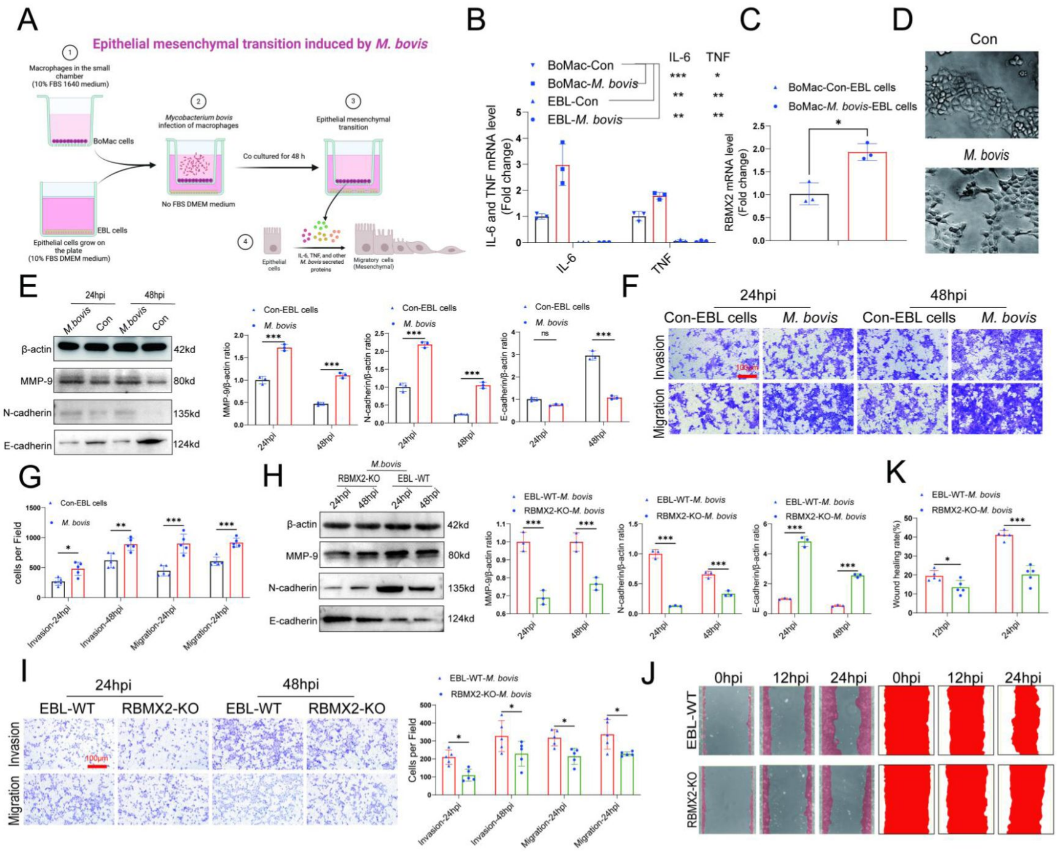


Figure 6.

RBMX2 facilitates EMT process in EBL cells after *M. bovis*-infected BoMac cells.

(A) A pattern diagram illustrated a *M. bovis*-infected BoMac cells inducing EMT of EBL cells coculture model, drawing by BioRender.
(B) Detection of IL-6 and TNF expression levels in EBL cells and BoMac cells infected with *M. bovis* using RT-qPCR. Data were relative to BoMac cells without infection of *M. bovis*.
(C) Detection of RBMX2 expression levels in a coculture model EBL cells after *M. bovis* infection using RT-qPCR. Data were relative to BoMac cells without infection of *M. bovis*.
(D) Observation of morphological changes in EBL cells infected with *M. bovis* under electron microscopy.
(E) EMT-related proteins (*MMP-9*, *N-cadherin*, and *E-cadherin*) expression was verified in coculture model EBL cells after *M. bovis* infection through WB. Data were relative to coculture model EBL cells without *M. bovis* infection
(F, G) The impact of coculture model EBL cells after *M. bovis* infection on migration and invasion capacity was detected using Transwell assay. Data were relative to coculture model EBL cells without *M. bovis* infection.
(H) The detection of EMT-related proteins (*MMP-9*, *N-cadherin*, and *E-cadherin*) of *RBMX2* knockout EBL cells after *M. bovis-infected* BoMac cells via WB. Data were relative to WT EBL cells after *M. bovis-infected* BoMac cells.
(I) The change in the migratory and invasive capabilities of *RBMX2* knockout EBL cells after *M. bovis-infected* BoMac cells was assessed via Transwell assay. Data were relative to WT EBL cells after *M. bovis-infected* BoMac cells
(J, K) Validate the changes in migration abilities of *RBMX2* knockout EBL cells after *M. bovis-infected* BoMac cells through wound healing assay. Data were relative to WT EBL cells after *M. bovis-infected* BoMac cells. T-test and two-way ANOVA were used to determine the statistical significance of differences between different groups. Ns presents no significance, * presents $p < 0.05$, **presents $p < 0.01$, and *** presents $p < 0.001$ indicate statistically significant differences. Data were representative of at least three independent experiments.

In conclusion, we successfully established a coculture model involving *M. bovis*-infected BoMac cells that induce EMT in EBL cells, thereby demonstrating that the host factor *RBMX2* effectively promotes the EMT process in this context.

RBMX2 facilitates the EMT process via the *p65/MMP-9* pathway

EMT enhances the migration and invasion capabilities of tumor cells [35–37]. Both p65 and *MAPK/p38/JNK* pathways have been shown to regulate the EMT process through various mechanisms. [38–41]. To elucidate the precise regulatory role of *RBMX2* in the EMT process within the coculture model of EBL cells, we assessed the *MAPK* pathway and p65 protein levels via WB. Our findings revealed a significant reduction in the phosphorylation of *p65* and *MAPK/p38/JNK* in *RBMX2* knockout cells infected with *M. bovis*, compared to WT EBL cells (**Supplementary Fig. 9C**).

To identify the specific signaling pathways regulating the EMT phenotype involving p65 and MAPK/p38/JNK in the coculture model, RBMX2 knockout EBL cells were treated with PMA, Anisomycin, and ML141. Our results showed that p65 suppresses E-cadherin expression while enhancing MMP-9 expression (**Fig. 7A**). Activation of the MAPK/p38 pathway inhibits E-cadherin expression and promotes N-cadherin expression. In contrast, the MAPK/JNK pathway suppresses N-cadherin expression while enhancing the expression of both E-cadherin and MMP-9. Furthermore, a comprehensive analysis of the relationship between pathway activation and cellular migration and invasion revealed that activation of the p65 pathway increases the migratory and invasive abilities of EBL cells, as demonstrated by Transwell assays (**Figs. 7B** and **C**) and wound-healing assays (**Figs. 7D** and **E**).

Invasion and migration are critical stages in tumor development, influenced by *p65*-dependent factors such as matrix metalloproteinases, urokinase fibrinogen activators, and interleukin-8 [42–46]. To further substantiate the correlation between the p65 protein and the regulation of the EMT process in EBL cells post-*M. bovis* infection, we employed siRNA to inhibit p65 expression in WT EBL cells. This suppression was found to inhibit *MMP-9* expression, as demonstrated by WB (**Figs. 7F** and **G**).

To thoroughly verify the regulatory mechanism between RBMX2 and p65, we initiated our investigation by conducting an in-depth analysis of the RBMX2 promoter region to identify potential interactions with the transcription factor p65. Initially, we performed molecular docking simulations to predict the binding affinity and interaction patterns between RBMX2 and p65 proteins. These simulations revealed multiple amino acid residues within the RBMX2 protein that formed strong, stable interactions with p65. The docking analysis yielded a high docking score of 1978.643 (**Fig. 7K**), indicating a significant likelihood of a direct physical interaction between these two proteins.

To complement the protein-protein interaction analysis, we next investigated whether p65 could directly bind to the promoter region of the RBMX2 gene at the transcriptional level. Using the JASPAR database, a comprehensive resource for transcription factor binding profiles, we queried the RBMX2 promoter sequence for potential p65 binding sites. This analysis identified several putative binding motifs, suggesting that p65 may act as a transcriptional regulator of RBMX2 expression.

To experimentally validate this transcriptional regulatory relationship, we employed a dual-luciferase reporter assay. We cloned the RBMX2 promoter region containing the predicted p65 binding sites into a luciferase reporter plasmid. This construct was then co-transfected into cultured cells along with a plasmid expressing p65. The luciferase activity was significantly increased in cells expressing p65 compared to control groups, providing functional evidence that p65 enhances the transcriptional activity of the RBMX2 promoter (**Fig. 7I**).

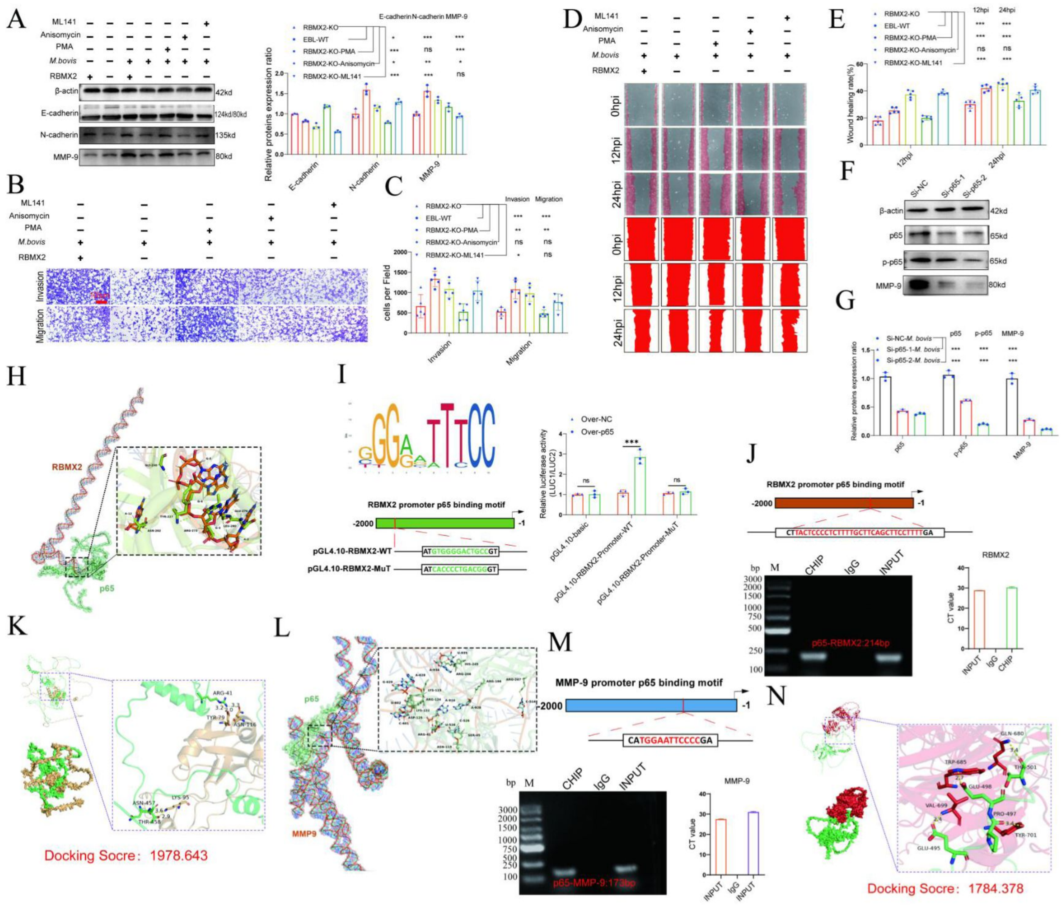


Figure 7.

RBMX2 facilitates the EMT in EBL cells via *p65/MMP-9* pathway.

(A) Evaluate the impact of pathway activations on expression of EMT-associated proteins (*MMP-9*, *N-cadherin*, and *E-cadherin*) in *RBMX2* knockout EBL cells after *M. bovis-infected* BoMac cells. Data were relative to *RBMX2* knockout EBL cells untreated activators

(B, C) Evaluate the impact of pathway activations on the migratory and invasive capabilities of RBMX2 knockout EBL cells after *M. bovis-infected* BoMac cells via Transwell assay. Data were relative to *RBMX2* knockout EBL cells untreated activators

(D, E) Evaluate the impact of pathway activations on the migratory capabilities of *RBMX2* knockout EBL cells after *M. bovis-infected* BoMac cells via wound healing assay. Data were relative to *RBMX2* knockout EBL cells untreated activators

(F, G) The impact of p65 silencing on the expression of *MMP-9* protein in WT EBL cells after *M. bovis* infection was assessed using WB. Data were relative to SiRNA-NC in WT EBL cells with *M. bovis* infection.

(H) Predicting the binding ability of RBMX2 promoter and p65 using molecular docking dynamics

(I) Verification of *RBMX2* promoter region and p65 interaction using dual luciferase reporter system.

(I) Using p65 antibody to precipitate p65 protein in EBL cells, and verification of *RBMX2* promoter region and p65 interaction using Chip-PCR assay.

(K) Predicting potential binding sites for p65 and *RBMX2* via protein docking.

(L) Verification of MMP-9 promoter region and p65 interaction using dual luciferase reporter system.

(M) Verification of MMP-9 promoter region and p65 interaction using Chip-PCR.

(N) Predicting potential binding sites for p65 and *MMP-9* via protein docking.

Two-way ANOVA was used to determine the statistical significance of differences between different groups. *p < 0.05, **p < 0.01, and ***p < 0.001 indicate statistically significant differences. Data were representative of at least three independent experiments.

Furthermore, to confirm the direct binding of p65 to the RBMX2 promoter in a chromatin context, we performed chromatin immunoprecipitation followed by quantitative PCR (ChIP-qPCR). In this assay, we used specific antibodies against p65 to immunoprecipitate chromatin fragments containing p65-bound DNA. The enriched DNA fragments were then analyzed using primers targeting the RBMX2 promoter region. Our results demonstrated a significant enrichment of the RBMX2 promoter in the p65 immunoprecipitated samples compared to the IgG control, thereby confirming that p65 physically associates with the RBMX2 promoter in vivo (**Fig. 7J**). Collectively, these findings-ranging from computational docking predictions to transcriptional reporter assays and ChIP validation-provide strong evidence supporting a direct regulatory interaction between p65 and RBMX2. This regulatory mechanism may play a critical role in the biological pathways involving these two molecules, particularly in contexts such as inflammation, immune response, or cellular stress, where p65 (a subunit of NF-κB) is known to be prominently involved.

Moreover, MMP-9 is known to induce EMT in epithelial cells, enhancing their invasiveness. The regulatory mechanisms involving p65 and *MMP-9* have been documented in several studies [47, 48]. In our research, we identified multiple binding sites in the promoter sequences of p65 and MMP-9 through molecular docking (**Fig. 7L**) and confirmed the biding site between *MMP-9* promoter and *p65* protein via Chip-PCR (**Fig. 7M**). Using protein docking, we validated the relationship between bovine p65 protein and MMP-9, with a binding score of 1784.378 (**Fig. 7N**).

In summary, *RBMX2* facilitates the EMT process of EBL cells following *M. bovis* infection by activating the *p65/MMP-9* pathway.

Discussion

The resurgence of *M. bovis*-associated TB presents a substantial global challenge, affecting both livestock and human populations. Notably, *M. bovis* exhibits over 99% nucleotide similarity with *M. tb* [49]. The pathogenesis of TB is complex process influenced by a confluence of bacterial, host, and environmental factors. Both *M. tb* and *M. bovis* have evolved sophisticated strategies to evade host immune responses, enabling their long-term intracellular persistence. It is estimated that approximately one-third of the global population harbors latent TB infections [50, 51].

The successful establishment of infection by mycobacteria is contingent upon their ability to circumvent early innate immune defenses, employing both transcriptional and post-transcriptional regulatory strategies within host macrophages [52–56]. Despite notable advancements in the field, unraveling the mechanisms of immune evasion and understanding the dynamics of latent infections caused by *M. tb* and *M. bovis* remains a substantial challenge. The cellular immune response to these pathogens is inherently complex, further compounded by the microdiversity of mycobacteria within individual hosts and the variability of immune responses among different individuals [54, 57].

Lung epithelial cells serve as the primary physical barrier against infection and play a pivotal role in the innate immune response. These cells not only function as a frontline defense but also recruit and activate antigen-presenting cells, such as macrophages, to initiate adaptive immune responses against *M. bovis* infection [58, 59]. Given that alveolar epithelial cells can act as reservoirs for *M. bovis*, their role in the infection process is particularly significant. Additionally, activated macrophages and neutrophils enhance the bactericidal effects of alveolar epithelial cells, further contributing to the host's defense mechanisms [60, 61].

In our study, we employed a CRISPR-Cas9 mutant library generated in our laboratory to explore potential host factors involved in M. bovis infection. Notably, we identified RBMX2 as a protein that significantly promotes M. bovis infection. The *RBMX2* protein, characterized by an RNA recognition motif within its 56-134 amino acid residue region, is implicated in mRNA splicing through spliceosomes and is emerging as a potential molecular marker for sperm activity.[62]. The downregulation of RBMX2 in the X chromosome of lung telocytes suggests its involvement in cellular immunity [63].

Further exploration of RBM genes in cattle revealed that EIF3G, RBM14, RBM42, RBMX44, RBM17, PUF60, SART3, and RBM25 belong to the same subfamily as RBMX2. Previous studies have demonstrated that genes within this subfamily can regulate the proliferation and lifecycle of cancer cells. For instance, EIF3G modulates the mTOR signaling pathway, inhibiting the proliferation and metastasis of bladder cancer cells [64]. Similarly, RBM14 has been linked to the reprogramming of glycolysis in lung cancer, acting as a novel epigenetically activated oncogene [60]. Despite these insights, the specific function of RBMX2 in the context of *M. bovis* infection and its potential role in cancer pathogenesis remain largely unexplored.

Our findings reveal that *RBMX2* is upregulated in TB-infected cells, demonstrating its capacity to enhance *M. bovis* infection. Transcriptomic analyses suggest that *RBMX2* may disrupt tight junctions within epithelial cells and promote EMT following *M. bovi*s infection.

Epithelial cells serve as more than passive barriers; they actively participate in innate immunity by regulating cytokine secretion and maintaining barrier integrity [61]. Disruptions in epithelial tight junctions, often exacerbated by pro-inflammatory stimuli from pathogenic bacteria, can facilitate bacterial translocation and subsequent infection [61, 65, 66]

In our experiments, we observed that the disruption of the epithelial barrier facilitated *M. bovis* adhesion and invasion. Conversely, the knockout of RBMX2 stabilized the epithelial barrier, attenuating *M. bovis* invasion and intracellular survival. This stabilization also mitigated downstream innate immune responses, reducing cellular inflammation, ROS production, and apoptosis.

The loss of tight junction integrity is a precursor to EMT, a process increasingly recognized for its role in cancer progression [20, 65–67]. Recent studies have established a link between bacterial infections and EMT induction, particularly in the context of gastric adenocarcinomas [68, 69]. Epidemiological evidence suggests that TB may serve as a risk factor for lung cancer [70, 71]; however, the underlying cellular mechanisms remain elusive. Chronic TB infection has been implicated in lung carcinogenesis, with reports indicating that BCG vaccination can enhance the survival of tumor cells under inflammatory conditions [72, 73]. There is growing epidemiological evidence suggesting that chronic TB infection represents a potential risk factor for the development of lung cancer. Studies have shown that individuals with a history of TB exhibit a significantly increased risk of lung cancer, particularly in areas of the lung with pre-existing fibrotic scars, indicating that chronic inflammation, tissue repair, and immune microenvironment remodeling may collectively contribute to malignant transformation [74]. Moreover, EMT not only endows epithelial cells with mesenchymal features that enhance migratory and invasive capacity but is also associated with the acquisition of cancer stem cell-like properties and therapeutic resistance [75]. Therefore, EMT may serve as a crucial molecular link connecting chronic TB infection with the malignant transformation of lung epithelial cells, warranting further investigation in the intersection of infection and tumorigenesis.

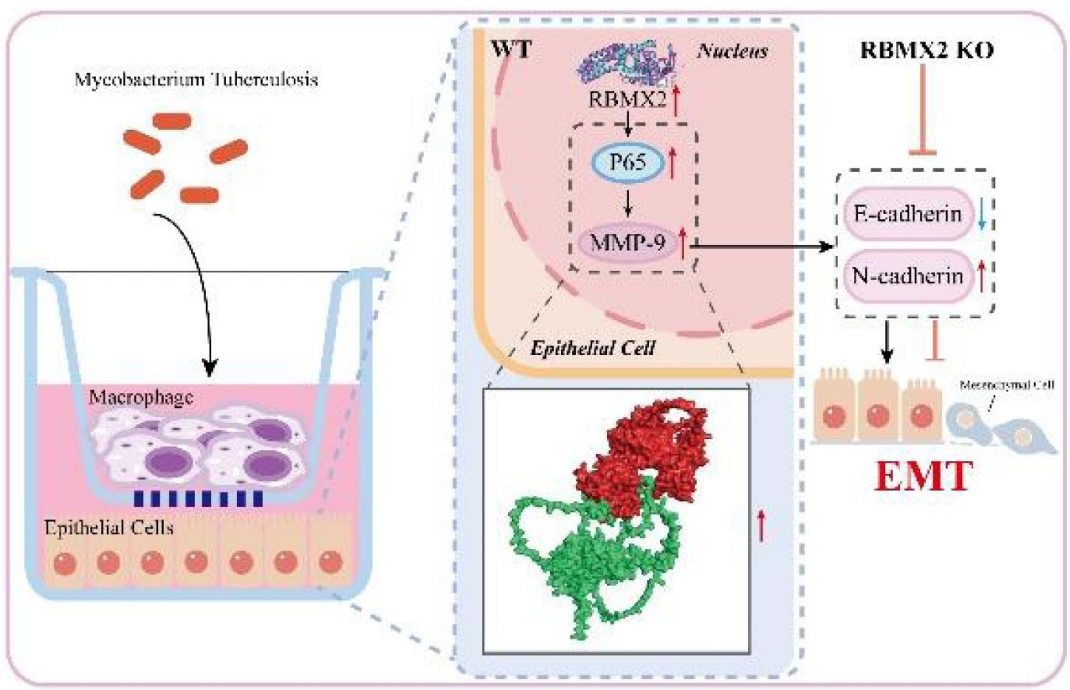
Moreover, *M. tb*-infected THP-1 cells have been shown to induce EMT in lung adenocarcinoma epithelial cells [76]. Chronic infection with *M. tb* is associated with oxidative stress and inflammatory cytokine production, fostering an environment conducive to tumor progression [77, 78]. Our analysis of RBMX2 across various cancers revealed increased expression levels in lung adenocarcinoma (LUAD) and lung squamous cell carcinoma (LUSC), suggesting a conserved role in tumor biology across species.

In light of these findings, we constructed a model of *M. bovis* infection in EBL cells to investigate EMT induction. Initial results indicated that *M. bovis* alone did not induce EMT; however, a co-culture model incorporating *M. bovis*-infected BoMac cells successfully induced EMT in EBL cells. Notably, the knockout of RBMX2 in this context inhibited EMT, suggesting that RBMX2 may elevate the risk of lung cancer through EMT induction following *M. bovis* infection. Meanwhile, metabolic pathways enriched after RBMX2 knockout, such as nucleotide metabolism, nucleotide sugar synthesis, and pentose interconversion, primarily support cell proliferation and migration during EMT by providing energy precursors, regulating glycosylation modifications, and maintaining redox balance; cofactor synthesis and amino sugar metabolism participate in EMT regulation through influencing metabolic remodeling and extracellular matrix interactions; chemokine and cGMP-PKG signaling pathways may further mediate inflammatory responses and cytoskeletal rearrangements, collectively promoting the EMT process.

In summary, RBMX2 drives TB pathogenesis by compromising epithelial barriers and inducing EMT. Targeting RBMX2 may present a promising avenue for the prevention and treatment of TB in both humans and animals. Additionally, the effective modulation of *RBMX2* could potentially mitigate the incidence of TB-associated EMT and its implications for lung cancer development in the near future.

Conclusion

Our research findings indicate that RBMX2 significantly enhances the invasive capacity of Mycobacterium bovis by promoting the activation and nuclear translocation of the p65 protein. This activation compromises the integrity of the epithelial cell barrier and induces epithelial-mesenchymal transition (EMT) through the p65/MMP-9 signaling pathway. These results highlight the crucial role of RBMX2 in M. bovis pathogenesis and underscore its potential as a therapeutic target for preventing infection-related complications.



Methods

Patients and lung tissues

This study was approved by the Ethics Committee of Inner Mongolia Autonomous Region People's Hospital, and written informed consent was obtained from all participating patients (Approval Number: No. 2020021). A total of 35 specimens of lung cancer tissue, along with adjacent normal lung tissue, were collected. We randomly selected the above specimens, including age and gender. All patients involved in this study had not received any medication, chemotherapy, or radiation therapy prior to surgical resection. The samples were subsequently prepared for immunohistochemistry (IHC) and fluorescence in situ hybridization (FISH) assays.

Bovine clinical tissues

This project has received approval from the Experimental Animal Ethics Center of Huazhong Agricultural University (Approval Number: HZAUCA-2024-0014). Following examination by researchers at the National Animal Tuberculosis Professional Laboratory (Wuhan), bovine tuberculosis tissue samples were collected from the lungs and livers of slaughtered cattle, with each specimen measuring at least 5 cm × 5 cm × 5 cm and specifically including tissues exhibiting tuberculous lesions.

Cell lines

EBL and 293T RRID:CVCL_0063 cells were generously provided by M. Heller from Friedrich-Loeffler-Institute. These cells were cultured in heat-inactivated 10% fetal bovine serum supplemented with Dulbecco's modified Eagle medium (DMEM, Gibco, USA) at 37°C and 5% CO_2 [79]. BoMac cells were kindly provided by Judith R. Stabel from the Johne's Disease Research Project at the United States Department of Agriculture in Ames, Iowa, and were maintained according to previously established protocols [80]. BoMac cells were grown in heat-inactivated 10% fetal bovine serum supplemented with Roswell Park Memorial Institute 1640 (RPMI 1640, Gibco, USA) at 37°C and 5% CO_2. Bovine lung alveolar primary cells were isolated in our laboratory and cultured in heat-inactivated 10% fetal bovine serum supplemented with DMEM (Gibco, USA) at 37°C and 5% CO_2. BEAS-2B RRID:CVCL_0168, NCIH1299, NCIH460 RRID:CVCL_0459, and CALU1 epithelial cells were all provided by Professor Shi Lei from Lanzhou University, and all culture procedures were conducted according to standardized protocols. We conduct regular testing on all cell lines to eliminate contamination from mycoplasma and black fungus.

Bacterium

M. bovis (ATCC:19210), originally isolated from a cow, is maintained in this laboratory. *M. smegmatis* mc^{2}155 (NC_008596.1) and *M. bovis* BCG-Pasteur (ATCC:35734) were generously provided by Professor Luiz Bermudez from Oregon State University. All strains were cultured in a Middlebrook 7H9 broth (BD, MD, USA) supplemented with 0.5% glycerol (Sigma, MO, USA), 10% oleic acid–albumin–dextrose–catalase (OADC, BD, USA) and 0.05% Tween 80 (Sigma, MO, USA) or on Middlebrook 7H11 agar plates (BD, MD, USA) containing 0.5% glycerol (Sigma, MO, USA) and 10% OADC (BD).

Prior to infection, the optical densities of bacterial cultures at 600 nm (OD600) were adjusted to the required MOI using the standard turbidimetric card. The cultures were then centrifuged at 3,000 g for 10 min. The precipitated bacteria were resuspended in medium and dispersed using an insulin syringe. Subsequently, 50 µL of 10-fold serially diluted bacterial suspension was plated onto Middlebrook 7H11 agar to determine the number of viable bacteria (colony-forming units, CFUs).

All experiments involving *M. bovis* were conducted in strict accordance with the biosafety protocols established for the Animal Biosafety Level 3 Laboratory of the National Key Laboratory of Agricultural Microbiology at Huazhong Agricultural University.

E. coli (ATCC:25922) was donated by Professors Zhou Rui and Wang Xiangru of Huazhong Agricultural University, while *Salmonella* (ATCC:14028) has been preserved and passed down through generations in our laboratory. These strains were resuscitated, and single colonies were purified, cultured, and grown in Luria–Bertani broth.

Generation of the *RBMX2*-KO EBL cells

The small guide RNA (sgRNA) sequence targeting the bovine RBMX2 gene (5 ’-GAATGAGCGTGAGGTCGAAC-3 ’) was cloned into the lentiviral vector pKLV2-U6gRNA5(BbsI)-PGKpuro2ABFP (#67991), kindly provided by Professor Zhao Shuhong from Huazhong Agricultural University, to generate the recombinant lentivirus. EBL cells were infected with either the RBMX2-targeting lentivirus or the empty vector lentivirus as a negative control. At 48 to 60 hours post-infection (hpi), puromycin (2.0 mg/mL) was added to select for stably transduced cells. Positive clones were subsequently enriched and monoclonal cell lines were obtained through limiting dilution. Successful knockout of RBMX2 was confirmed by PCR analysis..Table S1

Extraction of total RNA and RT-qPCR

Cold phosphate-buffered saline (PBS, HyClone, China) was used to wash the cells three times, after which 1 mL of Trizol (Invitrogen, USA) was added per well to lyse the cells. The lysate was collected in EP tubes, and 200 μL of chloroform was subsequently added. The mixture was vortexed for 30 seconds and then centrifuged at 12,000 rpm for 10 minutes at 4°C. Following centrifugation, 500 μL of the supernatant was transferred to a new EP tube. To this supernatant, 500 μL of isopropanol was added and mixed gently by inversion. The mixture was allowed to stand for 10 minutes at 4°C before being centrifuged again at 12,000 rpm for 15 minutes at 4°C. The supernatant was discarded, revealing the RNA pellet. Next, the RNA pellet was washed with 1 mL of 75% ethanol and centrifuged at 7,500 rpm for 5 minutes at 4°C. The supernatant was removed, and the RNA pellet was air-dried for 15 minutes. Subsequently, 20 μL of DEPC-treated water was added, and the mixture was incubated at 58°C in a water bath for 10 minutes to dissolve the RNA. Purified RNA was then obtained.

To assess RNA purity, the OD260/OD280 ratio was measured using a NanoDrop ND-1000 instrument (Agilent Inc., USA), with values between 1.8 and 2.0 indicating acceptable purity. RNA integrity and potential contamination with genomic DNA were evaluated using denaturing agarose gel electrophoresis. The samples were stored at −80°C for later analysis.

Reverse transcription of the RNA samples was performed using HiScript III RT SuperMix for qPCR (+gDNA wiper, Vazyme, China). Four microliters of 4× gDNA wiper mix were added to 1 μg of RNA, followed by the addition of 16 μL of RNase-free ddH2O. The mixture was incubated at 42°C for 2 minutes to eliminate genomic DNA. Reverse transcription was subsequently carried out by adding 4 μL of 5× HiScript III qRT SuperMix, with the mixture incubated at 37°C for 15 minutes and then at 85°C for 5 seconds to synthesize cDNA.

cDNA expression across different sample groups was quantified using AceQ qPCR SYBR Green Master Mix (Vazyme, China) in a ViiA7 real-time PCR machine (Applied Biosystems Inc., USA). The final volume of each real-time PCR reaction was 20 μL, comprising 10 μL of 2× AceQ qPCR SYBR Green Master Mix, 0.4 μL of upstream primer (10 μM), 0.4 μL of downstream primer (10 μM), 0.4 μL of 50× ROX reference dye 2, 3 μL of cDNA template, and 5.8 μL of ddH_2O. The PCR conditions were as follows: 95°C for 5 minutes (1 cycle), 95°C for 10 seconds (40 cycles), and 60°C for 30 seconds (40 cycles). The primer sequences for RT-qPCR are provided in **Table S2**.

sgRNA	sequence
sgRNA- F	5′-CCTTGCCCAATTTTTCGGAGG-3′
sgRNA- R	5′-ACAGGAGGATGGTAGTAACGG-3′

Table 1

Number	Primer	Sequence
1	Bovine-β-actin-F	5′-AGCAAGCAGGAGTACGATGAG-3′
2	Bovine-β-actin-R	5′-ATCCAACCGACTGCTGTCA-3′
3	Human-β-actin -F	5′- CACCATTGGCAATGAGCGGTTC-3′
4	Human-β-actin -R	5′- AGGTCTTTGCGGATGTCCACGT-3′
5	Bovine-RBMX2-F	5′-TCCCACCGGCACAAAAAGTC-3′
6	Bovine-RBMX2-R	5′-TCAGTGATACCTGAGCTCCC-3′
7	Human-RBMX2-F	5′-ACCCCCTCACCAAGTTTGTC-3′
8	Human-RBMX2-R	5′-GTGCTTCTTAGGGCCAGTGT-3′
9	Bovine-HSPB8-F-1	5′-GACCAAGGACGGATACGTGG-3′
10	Bovine-HSPB8-R-1	5′-AGTTCTTGGAGACGATGCCC-3′
11	Bovine-ST14-F-1	5′-CTCGGCGCTGGATTCAAGTA-3′
12	Bovine-ST14-R-1	5′-CAAGAACTCCACGCCTTCCT-3′
13	Bovine-MUC16-F-1	5′-TGTGCATGTGACAGCAGAGC-3′
14	Bovine-MUC16-R-1	5′-CTGGAAACTGGGAGTGGTCG-3′
15	Bovine-TNC-F-1	5′-TGGCATCGGAGAATGCCTTT-3′

Table 2

16	Bovine-TNC-R-1	5′-CAGCTTCCTCTGGGTTCCTG-3′
17	Bovine-GPX2-F-1	5′-ACCCGTTTTCCCTCATGACC-3′
18	Bovine-GPX2-R-1	5′-CGGCCCAATGAGGAACTTCT-3′
19	Bovine-THBS2-F-1	5′-GGGGACGCTTGTAAAGACGA-3′
20	Bovine-THBS2-R-1	5′-AATCTGAGTGGTGCCCTTGG-3′
21	Bovine-IL-7-F-1	5′-GCAATTGCCTGAATAACGAACCT-3′
22	Bovine-IL-7-R-1	5′-TTACCCTTGCTGGTGCAGTT-3′
23	Bovine-IL-24-F-1	5′-GACGCCTGCCCAACTTTCTT-3′
24	Bovine-IL-24-R-1	5′-AAGGGATGCTTGAGGGCTTG-3′
25	Bovine-FRMD4B-F-1	5′-CAGAGACAGGAGCACCAAGG-3′
26	Bovine-FRMD4B-R-1	5′-TTAAACTGCCGTTGCTTGCC-3′
27	Bovine-OMD-F-1	5′-GCTCAACGTTGGACACAACA-3′
28	Bovine-OMD-R-1	5′-AGCTTATTGGCGCTTTCAGC-3′
29	Bovine-NOS3-F-1	5′-GTTCCCTCGCGTGAAGAACT-3′
30	Bovine-NOS3-R-1	5′-CTGGTTGATGAAGTCCCTGGC-3′
31	Bovine-MCTP1-F-1	5′-AGTTCCGACTTGGGCATCAG-3′
32	Bovine-MCTP1-R-1	5′-TACTGAGAGCCGACAGGTCA-3′
33	Bovine-MFAP5-F-1	5′-TGATATCCAGCCTTCCACCG-3′
34	Bovine-MFAP5-R-1	5′-CCGATGCACGGAGTAGAGTC-3′
35	Bovine-MMP-9-1-F	5′-CGCTATGGCTACACTCCTGG-3′
36	Bovine-MMP-9-1-R	5′-TGAGTTCGCCCTCAAAGGTC-3′
37	Bovine-N-cadherin-1-F	5′-CACTGCGATTGATGCTGACG-3′
38	Bovine-N-cadherin-1-R	5′-TGCCACCGTGATAATGTCCC-3′
39	Bovine-E-cadherin 1-F	5′-CAGAAGATGACACCCGGGAC-3′
40	Bovine-E-cadherin 1-R	5′-AGCATCCAGGCCCCTATGTA-3′
41	Bovine-CLDN-5-F-1	5′-CCTGGACCACAACATCGTGA-3′
42	Bovine-CLDN-5-R-1	5′-GGAGTCGTACACCTTGCACT-3′
43	Bovine-OCLN-F-1	5′-CTTTCAAAAAGGGCTCCCGC-3′

Table 2 (continued)

44	Bovine-OCLN-R-1	5′-TGGATATTCCCTGATCCAGTCG-3′
45	Bovine-TJP1-F-1	5′-CTCAAGTTCCTGAAGCCCGT-3′
46	Bovine-TJP1-R-1	5′-TAGGATCACCCGACGAGGAG-3′
47	Bovine-TNF-F-1	5′-GAAGTTGCTTGTGCCTCAGC-3′
48	Bovine-TNF-R-1	5′-AACCAGAGGGCTGTTGATGG-3′
49	Bovine-IL-6-F-1	5′-CTTCTGCTTTCCCTACCCCG-3′
50	Bovine-IL-6-R-1	5′-TTCTGCCAGTGTCTCCTTGC-3′
51	Bovine-IL-1β-F-1	5′-AAAAGCTTCAGGCAGGTGGT-3′
52	Bovine-IL-1β-R-1	5′-AACTCGTCGGAGGACGTTTC-3′

Table 2 (continued)

Western blot

RIPA reagent (Sigma, USA), supplemented with protease inhibitors and phosphatase inhibitors (Roche, China), was added to the cell samples, and the cells were lysed on ice for 30 minutes. The supernatant was collected by centrifugation at 12,000 rpm and 4°C for 10 minutes. Protein concentrations of each sample were determined using a BCA kit (Beyotime, China), and the proteins were adjusted to equal concentrations.

The protein samples were mixed with 5× protein loading buffer and boiled for 10 minutes. Proteins were then separated using SDS-PAGE on 10% polyacrylamide gels at 100 V for 90 minutes, followed by transfer to polyvinylidene difluoride (PVDF) membranes (Millipore, Germany) at 100 V for 70 minutes. The PVDF membrane was incubated in 5% skimmed milk for 4 hours to block non-specific binding. The membrane was washed three times with Tris-buffered saline containing 0.15% Tween-20 (TBST) for 5 minutes each.

Subsequently, the membrane was incubated with a primary antibody at 4°C for 12 hours, followed by a secondary antibody at room temperature for 1 hour. The following primary antibodies were utilized: ZO-1 (Bioss, China), OCLN (Proteintech, China), CLADN-5 (Proteintech, China), MMP-9 (Proteintech, China), E-cadherin (Proteintech, China), N-cadherin (Bioss, China), LaminA/C (Proteintech, China), β-actin (Bioss, China), p65 (CST, USA), p-p65 (CST, USA), and a MAPK-related antibody (CST, USA). The following secondary antibodies were used: anti-mouse

HRP secondary antibody and anti-rabbit HRP (Invitrogen, USA)

Phylogenetic tree construction and gene structure and function predictive analysis of the bovine RBM family

The bovine RBM protein sequences were converted into FASTA format files and imported into MEGA-X for comparative analysis. The neighbor-joining method was employed to construct the phylogenetic tree of the CsGRF protein. The iTOL website (*https://itol.embl.de/*) was utilized to enhance the visualization of the CsGRF phylogenetic tree. Additionally, the MEME Suite database (*http://meme-suite.org/*) was used for the online analysis of conserved motifs within the RBM gene family.

Adhesion Assay

EBL cells were infected with different live bacteria at a multiplicity of infection (MOI) of 20 for durations of 15 minutes, 30 minutes, 1 hour, and 2 hours at 4°C. Following infection, the cells were washed three times with phosphate-buffered saline (PBS; Gibco, USA) to remove extracellular bacteria. After washing, cells were lysed using 0.1% Triton X-100 (Beyotime, China). The resulting lysates were plated onto 7H11 or LB agar plates and incubated for several days to enable enumeration of colony-forming units (CFUs).

Invasion Assay

EBL cells were infected with various live bacteria at an MOI of 20 for durations of 2, 4, and 6 hours at 37 °C. After infection, gentamicin (Procell, China) was added at a concentration of 100 μg/mL for 2 hours to eliminate extracellular bacteria. The infected cells were then washed three times with phosphate-buffered saline (PBS; Gibco, USA) to remove residual extracellular bacteria. Following thorough washing, the cells were lysed using 0.1% Triton X-100 (Beyotime, China). The resulting lysates were plated onto 7H11 or LB agar plates and incubated for several days to allow for colony-forming unit (CFU) enumeration.

Intracellular Survival Assay

EBL cells were infected with various live bacterial strains at a multiplicity of infection (MOI) of 20 for 6 hours at 37°C. After infection, extracellular bacteria were removed by treating the cells with gentamicin (100 µg/mL; Procell, China) for 2 hours. The infected cells were then washed three times with phosphate-buffered saline (PBS; Gibco, USA) to eliminate any remaining extracellular bacteria, marking the 0-hour time point.

Following this, the infected cells were cultured in complete medium supplemented with 0.1% Triton X-100 (Beyotime, China) and harvested at selected time points: 0, 24, 48, and 72 hours post-infection (hpi). Cell lysates were plated onto 7H11 or LB agar plates and incubated for several days to allow for colony-forming unit (CFU) enumeration.

RNA-Seq, proteomics and metabolomics

We collected cells at different time points after infection, where the cells at 0 hours were treated with gentamicin 2 hours post-infection with *M. bovis* to eliminate extracellular bacteria. Total RNA was isolated using Trizol Reagent (Invitrogen Life Technologies), and the concentration, quality, and integrity were determined using a NanoDrop spectrophotometer (Thermo Scientific). Three micrograms of RNA were used as input material for RNA sample preparation. Sequencing libraries were generated according to the following steps: mRNA was purified from total RNA using poly-T oligo-attached magnetic beads. Fragmentation was performed using divalent cations under elevated temperature in an Illumina proprietary fragmentation buffer. First-strand cDNA was synthesized using random oligonucleotides and Super Script II. Second-strand cDNA synthesis was subsequently performed using DNA Polymerase I and RNase H. The remaining overhangs were converted into blunt ends via exonuclease/polymerase activities, and the enzymes were removed. After adenylation of the 3' ends of the DNA fragments, Illumina PE adapter oligonucleotides were ligated for hybridization. To select cDNA fragments of the preferred length of 400–500 bp, the library fragments were purified using the AMPure XP system (Beckman Coulter, Beverly, CA, USA). DNA fragments with ligated adaptor molecules at both ends were selectively enriched using the Illumina PCR Primer Cocktail in a 15-cycle PCR reaction. The products were purified (AMPure XP system) and quantified using the Agilent high-sensitivity DNA assay on a Bioanalyzer 2100 system (Agilent). The sequencing library was then sequenced on the NovaSeq 6000 platform (Illumina) by Shanghai Personal Biotechnology Co., Ltd.

RNA sequencing services were provided by Personal Biotechnology Co., Ltd. Shanghai, China. The data were analyzed using the free online platform Personalbio GenesCloud (*https://www.genescloud.cn*). Proteomics and metabolomics were performed by METWARE.

Differential expression analysis of the transcriptome

The statistical methods of HTSeq (0.9.1) RRID:SCR_005514 were used to compare read count values for each gene, which served as raw measurements of gene expression, while FPKM (Fragments Per Kilobase of transcript per Million mapped reads) was used for data normalization. Subsequently, differential gene expression analysis was performed using DESeq (1.30.0) RRID:SCR_000154 under the following screening criteria: $|\text{log2FoldChange}| > 2$ indicated a significant fold change in expression, and a P-value < 0.05 was considered statistically significant. The resulting list of differentially expressed genes was used for further functional enrichment and pathway analysis.

For clustering analysis of differentially expressed genes, the Pheatmap (1.0.8) package in R was used to perform bidirectional clustering analysis on the expression patterns of all differentially expressed genes across various samples. The heatmap was constructed based on the expression levels of the same gene across different samples and the expression patterns of different genes within the same sample. Distance calculations were based on Euclidean distance, and complete linkage clustering was used to visualize overall expression differences among samples and similarities in gene expression patterns.

Moreover, to further enhance the flexibility and reproducibility of data analysis, a localized analysis workflow was introduced. All raw sequencing data (FASTQ files) underwent quality control (using FastQC) and were aligned (using STAR or HISAT2) to the reference genome, followed by gene expression quantification and differential analysis using locally installed HTSeq and DESeq2 pipelines. This local workflow not only improves the transparency and controllability of data processing but also enhances data privacy protection, making it particularly suitable for handling sensitive or large-scale datasets. All analysis parameters were thoroughly documented to ensure experimental reproducibility and result traceability.

Finally, data analysis was also supported by the free online analytical platform Personalbio GenesCloud (*https://www.genescloud.cn*), which provides a graphical interface and efficient cloud computing resources, supporting differential analysis, functional annotation (GO/KEGG), gene set enrichment analysis (GSEA), and visualization. The results from local and cloud-based analyses complement each other, ensuring comprehensive and accurate data interpretation.

GO and KEGG enrichment analysis

We mapped all genes to terms in the Gene Ontology (GO) database and calculated the number of differentially enriched genes for each term. GO enrichment analysis was conducted on the differentially expressed genes using the topGO package, where P-values were determined using the hypergeometric distribution method. A threshold of P-value < 0.05 was set to identify significant enrichment, allowing us to pinpoint GO terms associated with significantly enriched differential genes and to elucidate the primary biological functions they perform. Additionally, we employed ClusterProfiler (3.4.4) software to conduct KEGG pathway enrichment analysis for the differential genes, concentrating on pathways with a P-value < 0.05. The data analysis was carried out using the free online platform Personalbio GenesCloud (*https://www.genescloud.cn*).

Flow cytometry assay

Cells were washed with cold PBS and then centrifuged at 1,500 rpm for 5 minutes; the supernatant was discarded. The cells were fixed in 70% ethanol for 12 hours. Following fixation, the cells were centrifuged again at 1,500 rpm for 5 minutes, and the supernatant was removed. The cells were washed once with cold PBS.

For staining, a Cell Cycle and Apoptosis Analysis Kit (Beyotime, China) was utilized. The staining mixture was prepared with a total volume of 535 μL, comprising 500 μL of staining buffer, 25 μL of PI dye, and 10 μL of RNase A. The samples were incubated with the staining mixture at 37°C for 30 minutes prior to detection.

Cell viability assay

EBL cells were infected with *M. bovis* at an MOI of 100. Cell viability was assessed at 48, 72, 96, 120, and 144 hpi using the CCK-8 assay (Dojindo, Kumamoto, Japan).

At each time point, fresh medium containing 10% CCK-8 was added to a 96-well cell plate and incubated for 60 minutes at 37°C. The cell viability was then determined by measuring the absorbance at 450 nm. The percentage of cell viability was calculated using the following formula:

$$\text{cell viability (\%)} = [\text{A (infection group)} - \text{A (blank group)}] \,/\, [\text{A (infection group) A (blank group)}] \times 100$$

Wound-Healing Assay

A wound-healing assay was performed to evaluate the migratory capacity of treated cells following experimental intervention. Briefly, after a 48-hour co-culture of EBL cells with BoMac cells post-M. bovis infection, the cells were harvested and dissociated using 1x trypsin. Approximately 2 x 10^5 cells were then seeded into each well of a 6-well culture plate and allowed to adhere until they reached confluence and formed a uniform monolayer.

Once a confluent cell monolayer was achieved, a standardized “wound” was created by gently scratching the surface of the monolayer along a straight edge using a 200 μL pipette tip. To remove non-adherent and detached cells, the monolayer was rinsed three times with phosphate-buffered saline (PBS, Gibco, USA). Subsequently, serum-free culture medium was added to each well to minimize cell proliferation and to better assess cell migration.

Phase-contrast images of the wound area were captured at defined time points-0, 12, and 24 hours-using an inverted microscope. The wound closure rate was quantitatively analyzed using ImageJ software by measuring the change in wound width over time. The percentage of wound healing was calculated relative to the initial wound width at time 0.

This assay provided insights into the effects of M. bovis infection and subsequent treatments on cell migration, a critical component of the wound-healing process.

Cell Adhesion Assay

To evaluate the impact of RBMX2 knockout on the adhesive properties of EBL cells following M. bovis infection, a cell adhesion assay was performed using a commercially available cell adhesion detection kit (Bestbio, China), following the manufacturer’s instructions.

Matrigel (20 mg/L), used to simulate an extracellular matrix environment, was coated onto the wells of a 96-well plate at a volume of 100 μL per well. Subsequently, 3 x10^{4} cells (from either RBMX2 knockout or wild-type (WT) EBL cell lines) were seeded into each well and incubated for 6 hours in a CO_2 incubator at 37°C to allow cell attachment.

Following incubation, non-adherent cells were gently removed by washing the wells twice with phosphate-buffered saline (PBS). Adherent cells were then fixed and stained with 10 μL of cell stain solution A per well and incubated for an additional 2 hours at 37 °C.

The absorbance was measured at 560 nm using a microplate reader to quantify the intensity of staining, which correlates with the number of adhered cells. The results were used to compare the adhesive capacities of RBMX2 knockout versus wild-type EBL cells under the experimental conditions.

This assay provided a quantitative assessment of cell-matrix adhesion, offering insights into the potential role of RBMX2 in modulating cell adhesion during M. bovis infection.

Transwell Assay

To evaluate the migratory and invasive potential of treated EBL cells, a Transwell assay was performed using Transwell chambers (Corning, USA). A total of 5 x 10^4 cells suspended in 200 μL of serum-free culture medium were seeded into the upper chamber, while 550 μL of complete medium containing fetal bovine serum was added to the lower chamber to serve as a chemoattractant.

The cells were allowed to migrate through the membrane for 24 hours in a CO_2 incubator at 37 °C.

Following incubation, the Transwell insert was carefully removed, and the cells on the upper surface of the membrane were gently wiped away with a cotton swab. The migrated cells on the lower surface were then fixed by incubation with 550 μL of 4% paraformaldehyde (Beyotime, China) for 30 minutes at room temperature. After fixation, the cells were stained with 550 μL of crystal violet solution (Beyotime, China) for an additional 30 minutes to allow for visualization of the migrated cells.

The stained membrane was rinsed briefly with PBS for 30 seconds to remove excess dye, air-dried upside down, and then carefully removed from the insert. It was subsequently mounted onto a glass slide and sealed with neutral balsam to preserve the staining.

Images were captured under a light microscope, and three random fields of view were selected for analysis. The number of migrated cells was quantified using ImageJ software (RRID:SCR_003070). The results were used to compare the migratory abilities of different experimental groups.

This assay provided a quantitative assessment of cell migration, which is essential for understanding the functional impact of treatments or genetic modifications on cell motility.

Crystal violet staining assay

EBL cells were infected with *M. bovis* at an MOI of 100. The surviving cells on six-well plates were then fixed with 4% paraformaldehyde (Beyotime, China). After fixation, the cells were stained with 0.1% crystal violet staining solution (Beyotime, China) and rinsed five times with PBS (Gibco, USA). The plates were placed in a 37°C oven for 6 hours before being photographed.

M. bovis infection of BoMac and coculture with EBL cells

BoMac cells were seeded into the upper chamber of a 0.4 μm pore size Transwell (Corning, USA) at a density of 3×10^4 cells per insert in RPMI 1640 medium supplemented with 10% FBS. The following day, the cells were infected with *M. bovis* at an MOI of 5 for 4 hours. After infection, the cells were washed three times with 1× warm PBS and treated with Gentamycin (100 μg/mL) for 2 hours to eliminate any extracellular bacilli.

The infected and uninfected BoMac cells were then incubated for 24 hours to mitigate the effects of *M. bovis* infection. Afterward, the cells underwent an additional 24-hour incubation period in fresh RPMI 1640 medium before coculturing with EBL cells.

In parallel, EBL cells were seeded at a density of 2×10^5 cells per well onto 12-well plates containing DMEM supplemented with 10% FBS. The following day, culture inserts containing either *M. bovis*-infected or uninfected BoMac cells were introduced into the wells of the 12-well plates containing EBL cells, and the coculture was incubated for up to 24 and 48 hours in serum-free DMEM.

Small interfering RNA (siRNA) transfection

EBL cells were transfected with p65 siRNA (Tsingke, China) using jetPRIME (Polyplus, France) following the manufacturer's instructions. Scrambled siRNA (Tsingke, China) served as a negative control.

Cells were seeded in 12-well plates and cultured at 37°C in a 5% CO_2 atmosphere. To prepare the transfection mixture, 0.8 μg of jetPRIME was incubated with 1 μL of siRNA for 10 minutes. This mixture was then added to the cells and incubated for 12 hours.

Following the incubation, RNA extraction was performed, and the efficiency of transfection was calculated.

The siRNA sequences used were as follows:

Cattle p65-1 siRNA, sense 5'-GCAGUUUGAUACCGAUGAA (dT)(dT)-3';

Cattle p65-1 siRNA, antisense 5'-UUCAUCGGUAUCAAACUGC (dT)(dT)-3';

Cattle p65-2 siRNA, sense 5'-GGACGUACGAGACCUUCAA (dT)(dT)-3';

Cattle p65-2 siRNA, antisense 5'-UUGAAGGUCUCGUACGUCC (dT)(dT)-3'.

P65 nuclear translocation assay

For the p65 nuclear translocation assay, RBMX2 knockout and wild-type cell lines were transfected with the pCMV-EGFP-p65 plasmid. Following transfection, bovine lung epithelial cells were infected with *M. bovis*.

After infection, the cell nuclei were stained with Hoechst dye. The entry of p65 into the nucleus was analyzed at different time points using a confocal high-resolution cell imaging analysis system from PerkinElmer Life and Analytical Sciences Ltd. (Britain). This imaging technique allowed for the assessment of p65 translocation into the nucleus in response to infection.

CHIP-PCR

Formaldehyde cross-linking and ultrasonic fragmentation of cells were performed. One plate of cells was removed, and the volume of the culture medium was measured. 37% formaldehyde was added to achieve a final concentration of 1%, and the plates were incubated at 37°C for 10 minutes. Glycine (2.5 M) was then added to the plates at a final concentration of 125 mM to terminate cross-linking. After mixing, the culture medium was left at room temperature for 5 minutes, and the cells were cleaned three times with cold PBS. The cells were scraped off with PBS, centrifuged at 2,000g for 5 minutes, and the supernatant was removed. IP lysis solution containing a protease inhibitor was added to lyse the cells (the amount of lysis solution depending on the amount of cell precipitation), and the mixture was fully lysed on ice for 30 minutes, with the cells being repeatedly blown with a gun or shaken on a vortex mixer to ensure complete lysis.

After sonication, the mixture was centrifuged at 12,000 rpm at 4°C for 10 minutes to remove insoluble substances and collect the supernatant. 90 μL of the input was retained, and the rest was stored at −80°C. To confirm the presence of the target protein in the sample, 40 μL of the ultrasonic crushing product was taken, mixed with 10 μL of L5 reduced protein loading buffer, and heated for denaturation before performing Western blot detection.

For the remaining 50 μL of the product, 5 μL of Protease K and 2 μL of 5M NaCl (final NaCl concentration of 0.2M) were added and incubated at 55°C overnight for cross-linking. After cross-linking, the nucleic acid concentration was measured, and a portion of the sample was taken for PCR amplification, followed by agarose gel electrophoresis to detect the effectiveness of ultrasonic fragmentation and confirm the presence of the target DNA. After verifying the Input result, 100 μL of the ultrasonic crushing products was frozen at −80°C.

Next, 900 μL of ChIP Diffusion Buffer containing 1 mM PMSF and 20 μL of 50% of 1× PIC was added, along with an additional 60 μL of Protein A+G Agarose/Salmon Sperm DNA. The mixture was stirred at 4°C for 1 hour, allowed to stand at 4°C for 10 minutes to precipitate, and then centrifuged at 4000 rpm for 5 minutes. The sample was divided into two 1.5 mL EP tubes; the target protein IP antibody was added to one tube, while IgG (1 μg of the corresponding species) was added to the other. The samples were shaken overnight at 4°C for precipitation and cleaning of immune complexes.

After the overnight incubation, 200 μL of Protein A+G Agarose/Salmon Sperm DNA was added to each tube, shaken at 4°C for 2 hours, allowed to stand at 4°C for 10 minutes, and then centrifuged at 4,000 rpm for 1 minute. The supernatant was removed, and 8 μL of 5M NaCl and 20 μL of Protein K were added for overnight cross-linking at 55°C. Using databases like JASPAR to predict transcription factor binding sites, primers were designed and synthesized based on these binding sites. RT-PCR was employed to verify the binding, and after amplification, the products were taken for gel electrophoresis to confirm the correct fragment size (**Table S3**).

Table 3

Gene name	Sequence
MMP9	5′-AGGCACTGAAGGAATGAGGC-3′
	5′-CCCCGCTTGGTGGAAAT-3′
RBMX2	5′-CTGACAGGAGGACCAGAGTTT-3′
	5′-TGCCTAGTTGTAGGGTTGAAGT-3′

Dual-luciferase reporter assay

293T cells in the logarithmic growth phase were adjusted to a density of 5×10^4 cells/mL and inoculated into 48-well cell culture plates, with 300 cells per well. Each concentration gradient was represented by three replicate wells. On the second day after plating, when the cells reached a density of approximately 60-70%, transfection was performed with the following experimental groups: pGL4.10 RRID:Addgene_72684 basic+pRL TK+Over NC, pGL4.10 basic+pRL TK+Over p65, pGL4.10-*RBMX2*-WT+pRL TK+Over NC, pGL4.10-*RBMX2*-WT+pRL TK+Over p65, pGL4.10-*RBMX2*-MUT+pRL TK+Over NC, pGL4.10-*RBMX2*-MUT+pRL TK+Over p65. Transfection complexes were prepared, and dual-luciferase detection was carried out 48 hours post-transfection.

For cell lysis, the cell culture medium was removed, and the cells were gently rinsed twice with PBS. Then, 50 μL of 1× PLB lysis solution was added to each well, and the plates were placed on a shaking table at room temperature for 15 minutes to ensure complete lysis.

From each sample, 20 μL was taken for measurement. To assess Firefly luciferase activity, 100 μL of Firefly luciferase detection reagent (LAR II reagent, Promega, USA) was added, mixed well, and the relative light unit (RLU1) was measured. After this, 100 μL of Sea Kidney luciferase detection reagent (1× Stop&Glo® Reagent, Promega, USA) was added, mixed thoroughly, and the relative light unit (RLU2) was measured.

The activation level of the target reporter genes was compared between different samples by calculating the ratio of RLU1 from the Firefly luciferase assay to RLU2 from the Sea Kidney luciferase assay.

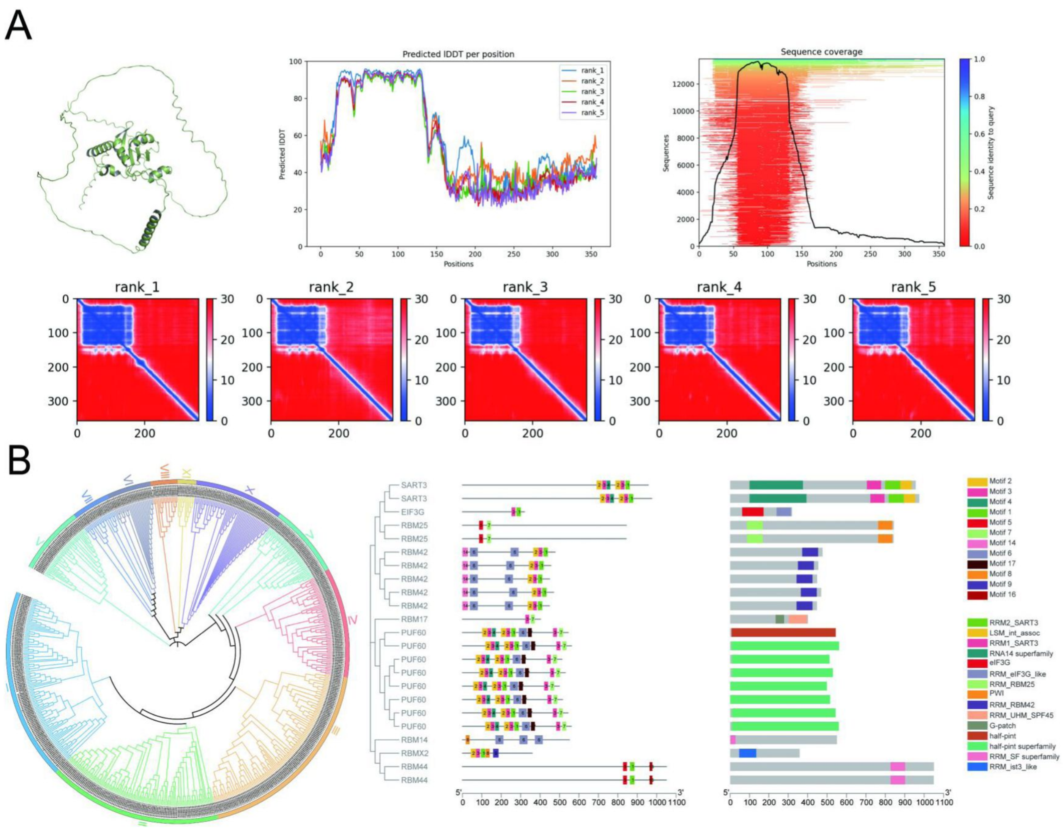
Protein docking

Molecular docking simulations were conducted to predict the formation of stable complexes between proteins RBMX2 or MMP9 with p65. The aligned sequences of RBMX2, MMP9, and p65 proteins were sourced from the UniProt RRID:SCR_002380 database. Their three-dimensional structures were predicted using AlphaFold and refined by constructing the structures in Avogadro. The structures were then optimized using the MMFF94 force field and exported in PDB format for further optimization in Gauss09.

The docking of the proteins was performed using Hdock, where each protein was treated separately as the receptor and ligand to assess their interactions [81]. The resulting docking affinities were annotated to provide insight into the binding interactions. Finally, PyMOL RRID:SCR_000305 was utilized to visualize the binding interaction geometries, allowing for a detailed examination of the molecular interactions between the proteins.

Statistical analysis

Statistical analysis was performed on all assays conducted in triplicate, with data expressed as the mean ± standard error of the mean. Each experiment was independently repeated three times. GraphPad Prism 7.0 RRID:SCR_002798 (La Jolla) was utilized for statistical analysis, employing a two-tailed unpaired t-test with Welch's correction for comparisons between two groups. For comparisons among multiple groups, one-way or two-way ANOVA was applied, followed by the LSD test. Statistical significance was indicated at four levels: not significant (ns presents $p > 0.05$), * presents $p < 0.05$, ** presents $p < 0.01$, and *** presents $p < 0.001$.

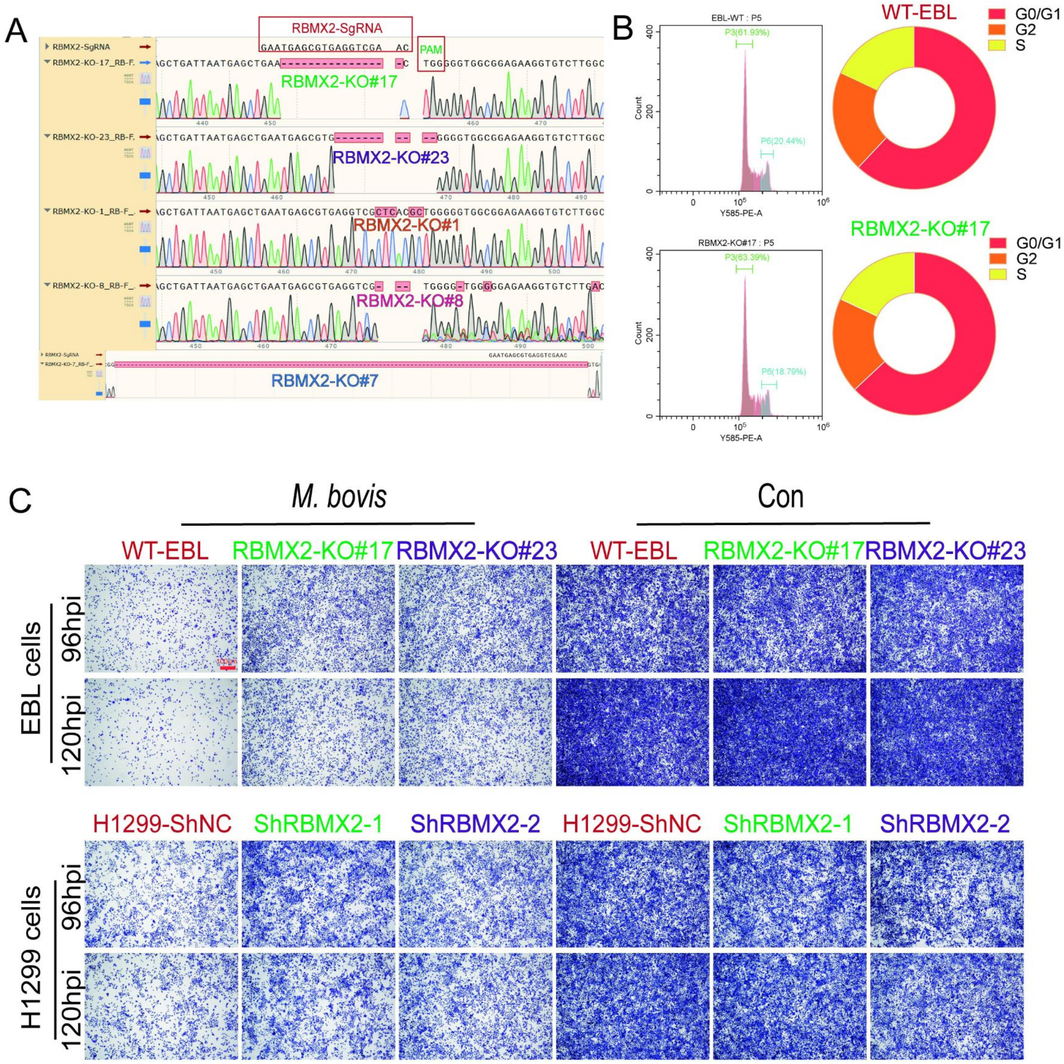


Supplementary Fig 1

Protein structure analysis, subfamily localization.

(A) AlphaFold Multimer predicted the sequence and structure of *RBMX2* protein.

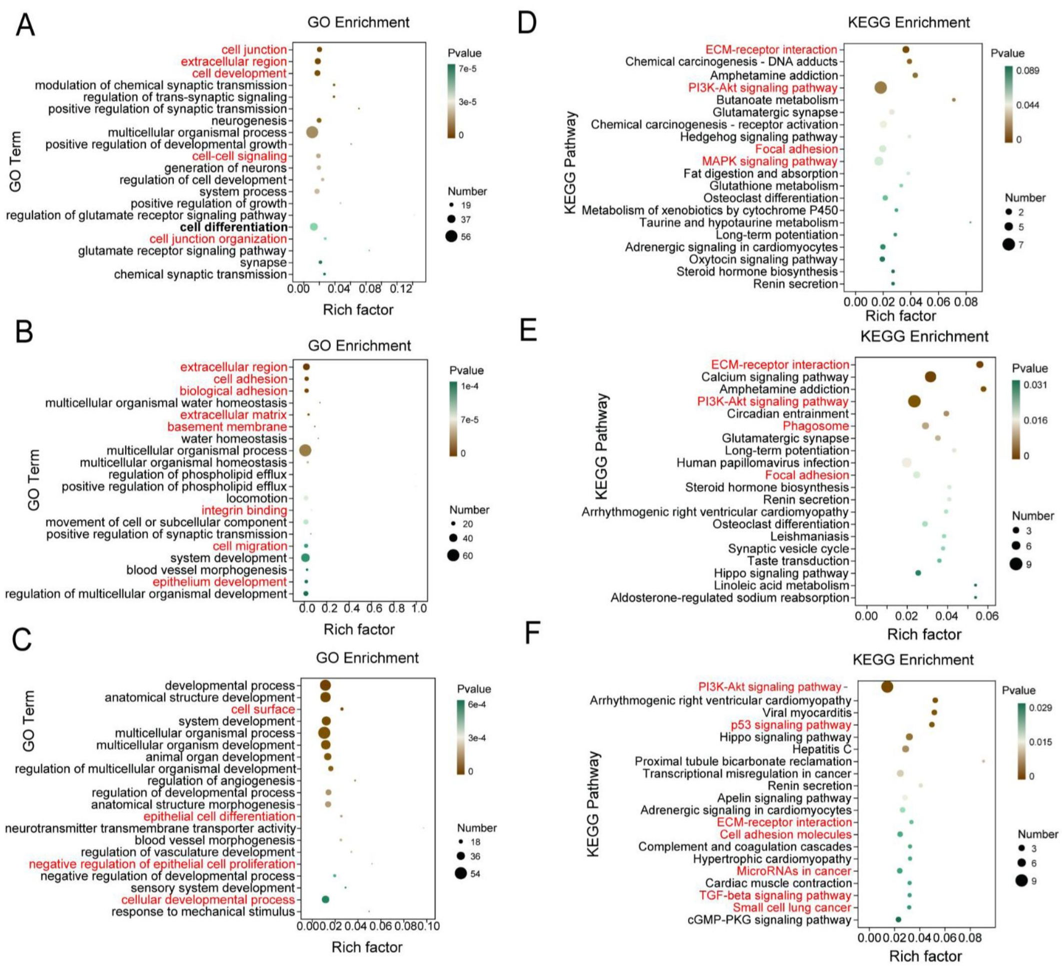
(B) Construction of RBM Gene Family Tree and structural Analysis of genes in *RBMX2* subfamily.



Supplementary Fig 2

The influence of *RBMX2* in cell cycle, cell morphology, cell proliferation, and resistance to *M. bovis* infection.

(A) The different knockout sites of *RBMX2* EBL cells were observed by sequencing compared to the bovine *RBMX2* sequence or WT EBL cells.
(B) Observation of the effect of *RBMX2* knockout on cell cycle of EBL cells by flow cytometry assay. Data are represented as the G0/G1 and S phase relative to WT EBL cells.
(C) The change in cell number of *RBMX2* knockout and silence following 96 and 120 hours of *M. bovis* infection was observed via crystal violet assay in EBL cells and H1299 cells, respectively. Data were represented as the cell number relative to WT EBL cells and H1299-ShNC cells after *M. bovis*-infection. Data were representative of at least three independent experiments.

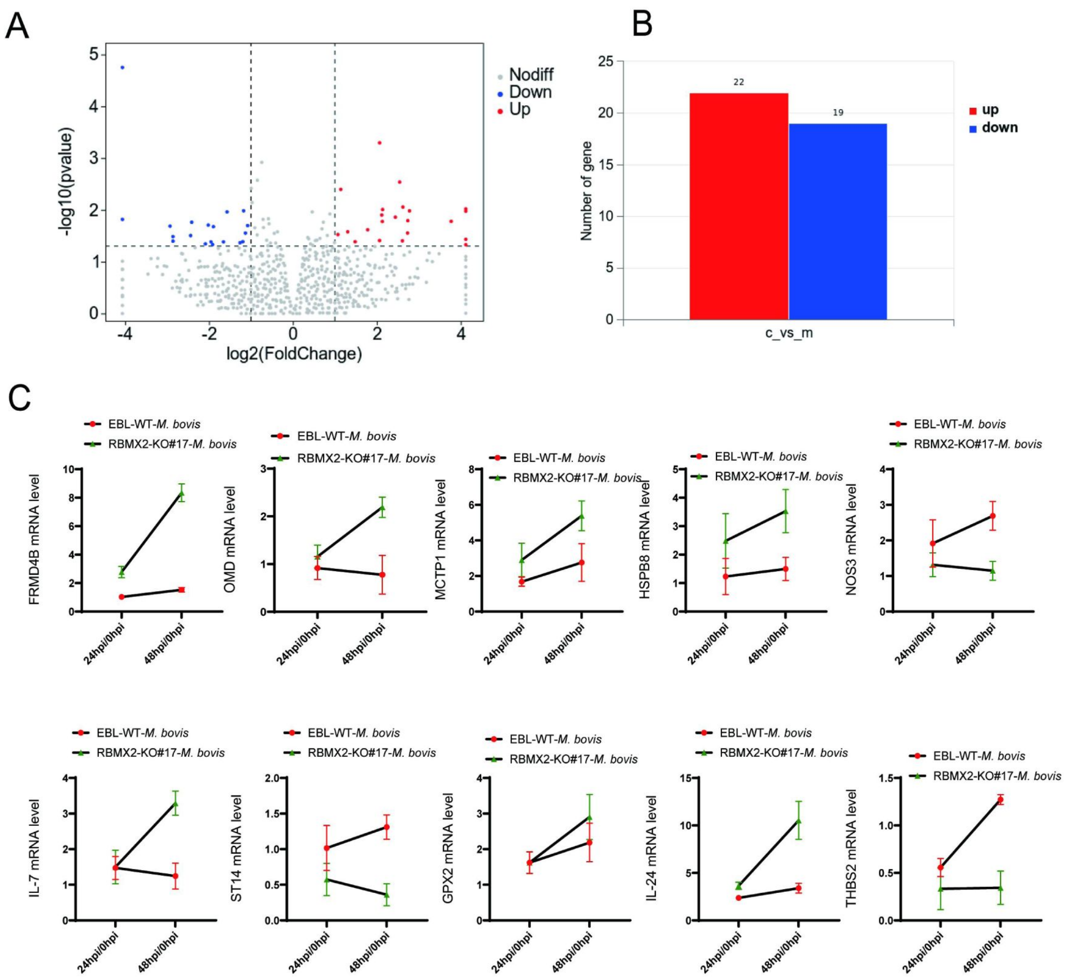


Supplementary Fig 3

Transcriptome analysis in *RBMX2* knockout and WT EBL cells after *M. bovis* infection in different time points

(A-C) GO analysis of enriched genes of (A) 0, (B) 24 and (C) 48h after *M. bovis* infection, respectively. Data were in RBMX2 knockout EBL cells relative to WT EBL cells with *M. bovis* infection. The changes of these pathways from cell junction-related pathways in 0 hpi to cell proliferation and differentiation-related pathways in 48 hpi.

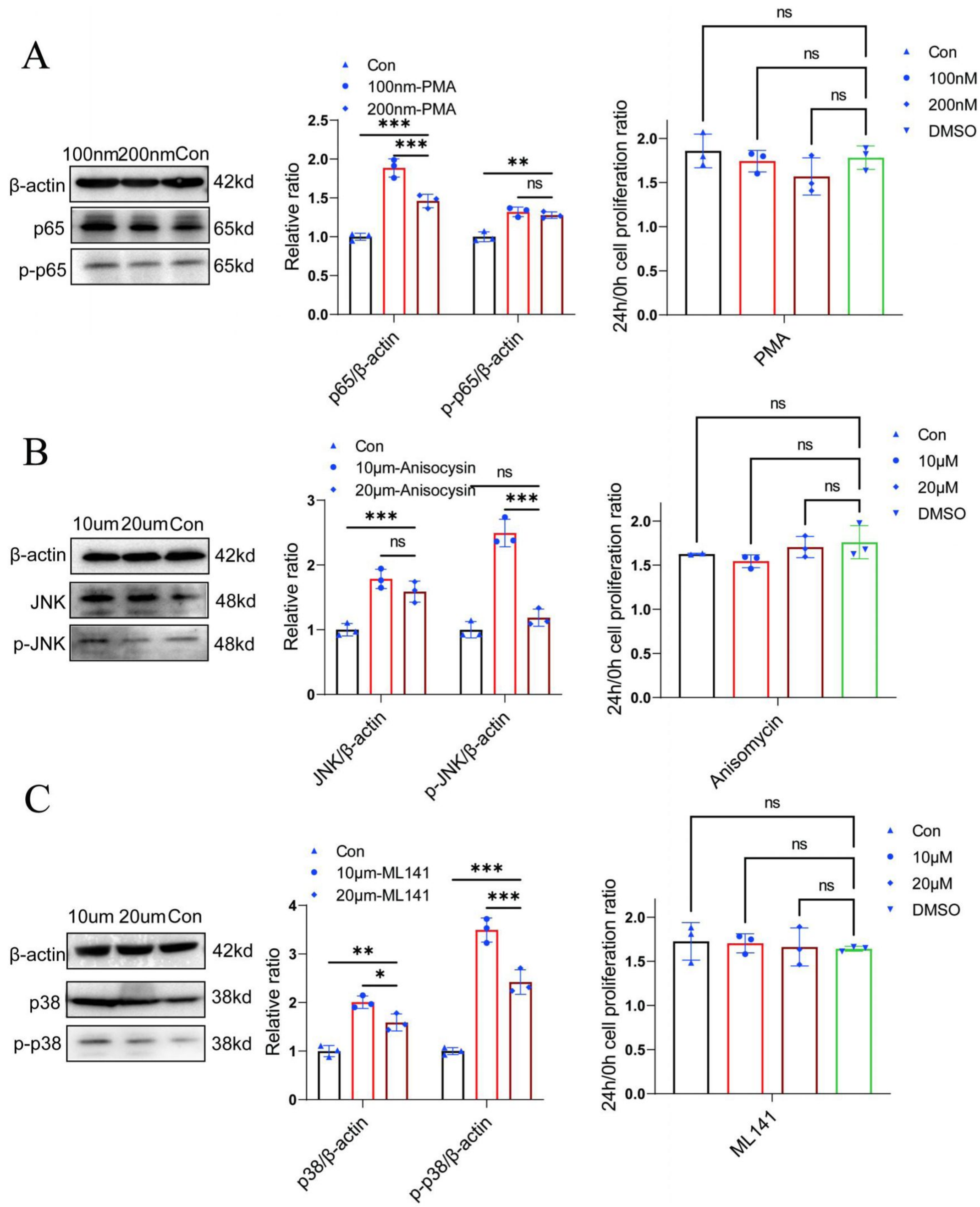
(D-F) KEGG analysis of enriched genes of (D) 0, (E) 24 and (F) 48hpi after *M. bovis* infection, respectively. Data were in RBMX2 knockout EBL cells relative to WT EBL cells with *M. bovis* infection. The changes of these pathways from inflammation-related pathways in 0 hpi to cancer-related pathways in 48 hpi.



Supplementary Fig 4

Transcriptomic analysis of *M. bovis* infected EBL cells.

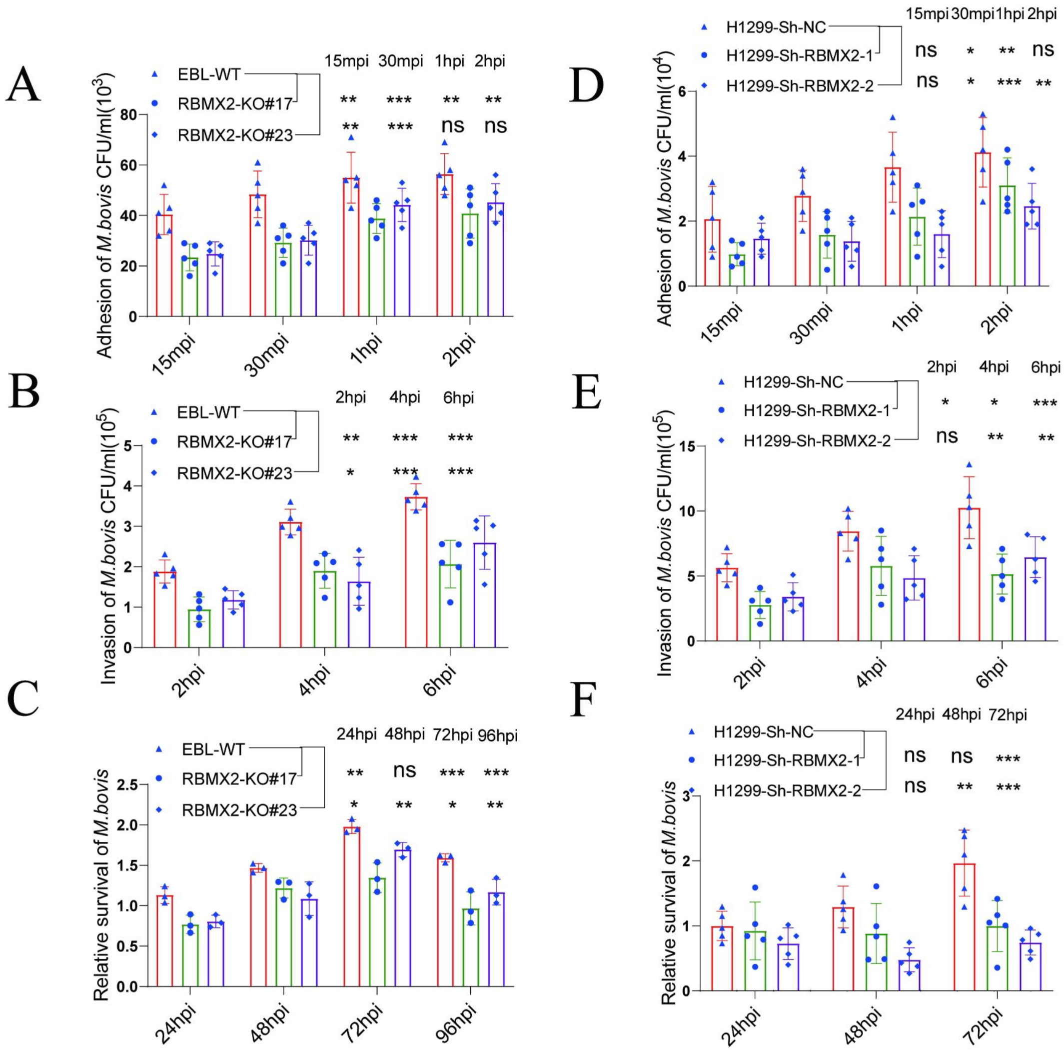
(A, B) A volcano map illustrating the transcriptional enrichment genes after WT EBL cells with *M. bovis*-infection. Data were relative to WT EBL cells without *M. bovis* infection.
(C)Identification the expression of related genes mRNA enriched by RT-qPCR. Data were represented as the fold expression in 24hpi/48hpi relative to 0hpi (2 hpi post-gentamicin, recorded as 0 hpi).



Supplementary Fig 5

The optimal concentration of activators and their impact on cell viability.

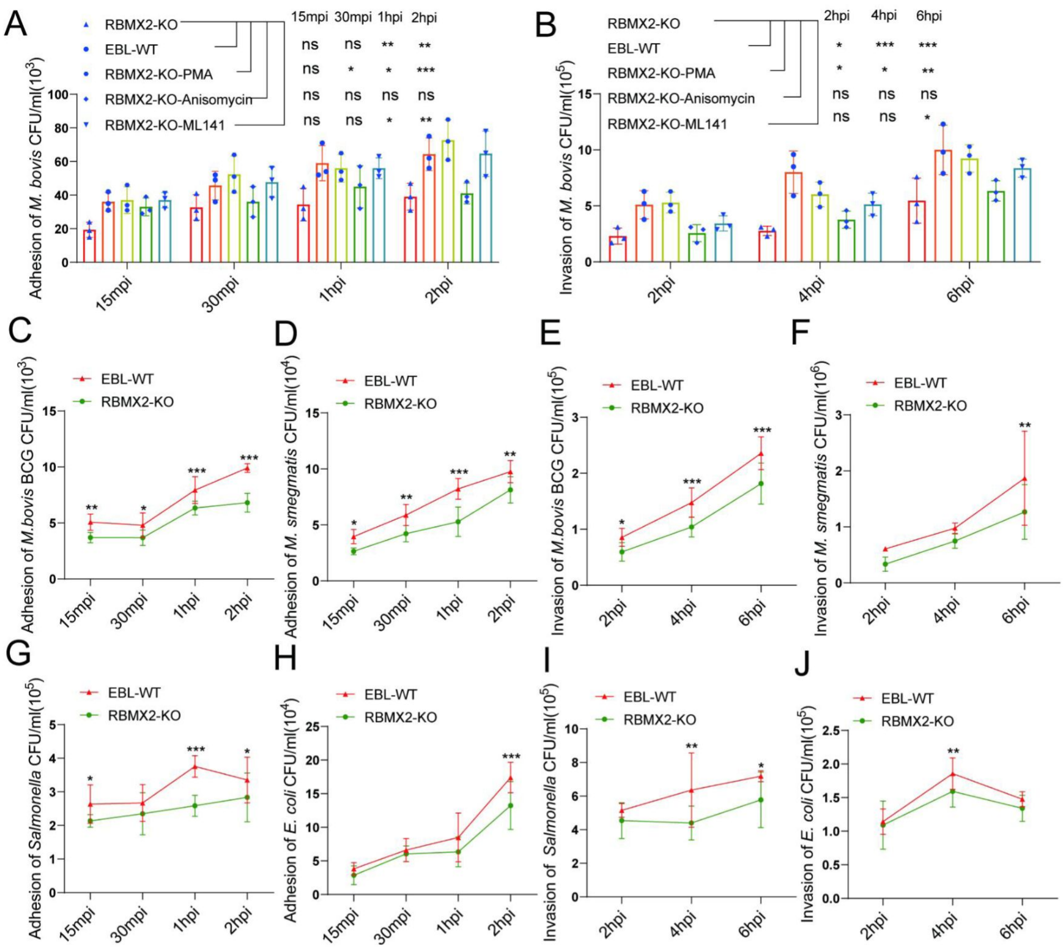
(A-C) Ascertain the optimal concentration of the three pathway activators and their impact on cellular viability via WB and CCK-8 assays. Data were relative to WT EBL cells untreated activators. One-way ANOVA was used to determine the statistical significance of differences between different groups. Ns presents no significance, **presents $p < 0.01$, and *** presents $p < 0.001$ indicate statistically significant differences.



Supplementary Fig 6

***RBMX2* enhanced the processes of *M. bovis* adhesion, invasion, and intracellular survival.**

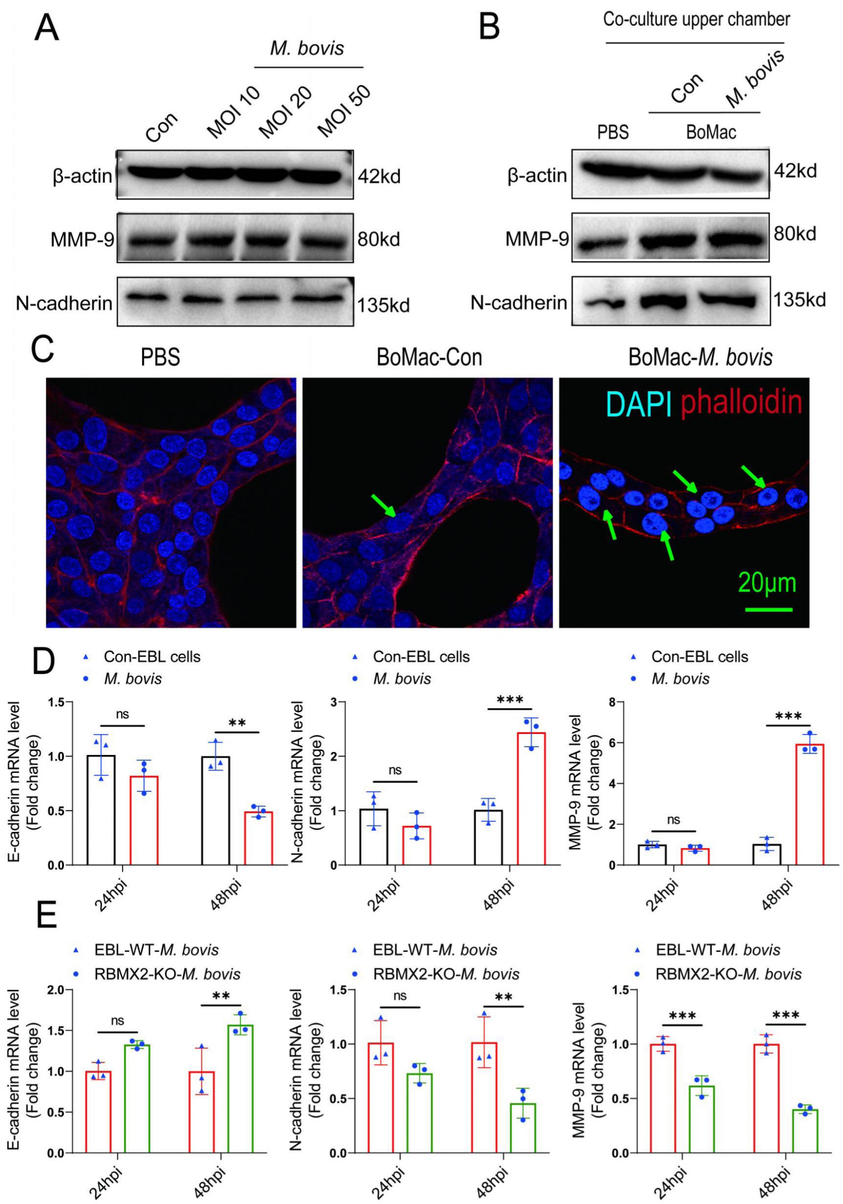
(A, B, C) The impact of *M. bovis* on the adhesion, invasion, and intracellular survival of *RBMX2* knockout and WT EBL cells through plate counting. Data were relative to WT EBL cells after *M. bovis* infection.
(D, E, F) The impact of *M. bovis* on the adhesion, invasion, and intracellular survival of Sh-NC and Sh-RBMX2 H1299 cells through plate counting. Data were relative to H1299 ShNC cells after *M. bovis* infection.
One-way ANOVA and two-way ANOVA were used to determine the statistical significance of differences between different groups. Ns presents no significance, * presents $p < 0.05$, **presents $p < 0.01$, and *** presents $p < 0.001$ indicate statistically significant differences. Data were representative of at least three independent experiments.



Supplementary Fig 7

***RBMX2* enhanced the processes of pathogen adhesion, invasion, and intracellular survival.**

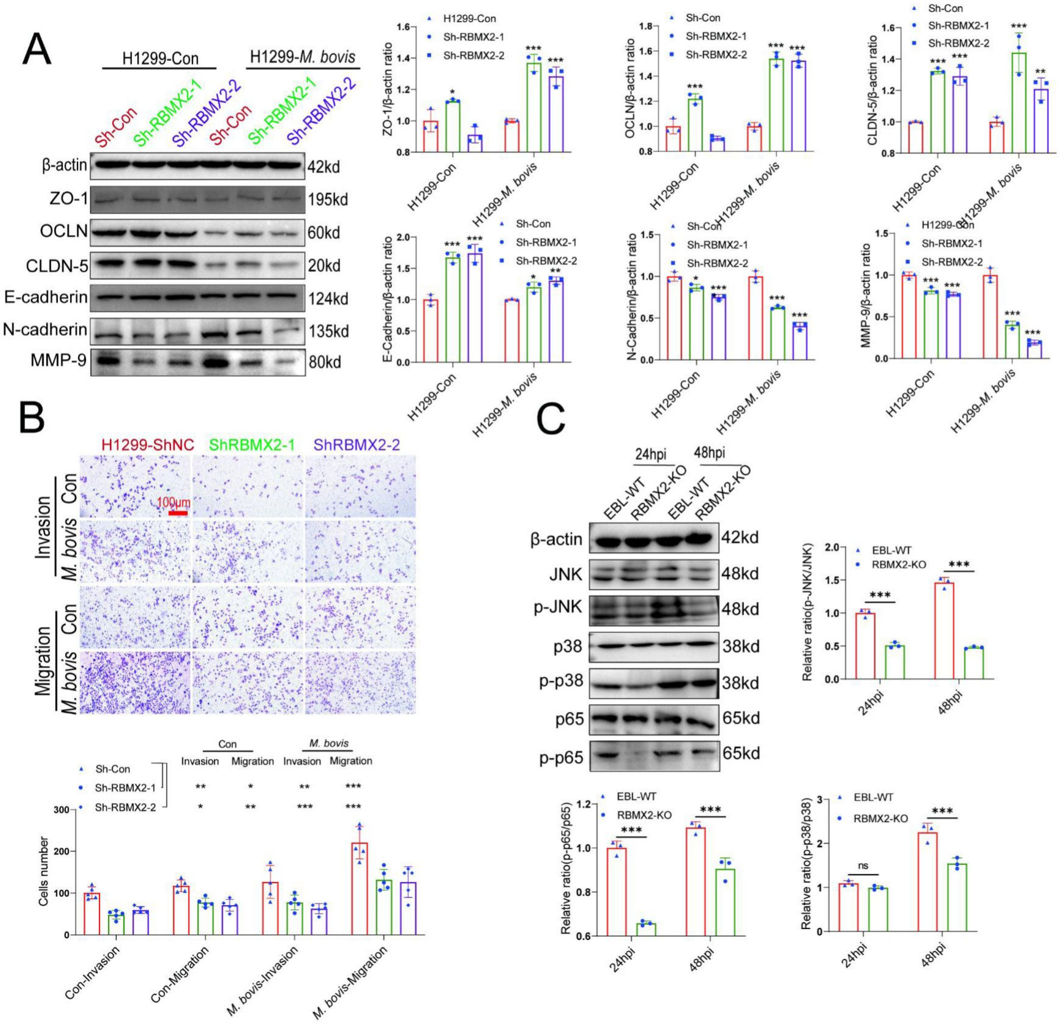
(A, B) The impact of *M. bovis* on the adhesion and invasion of *RBMX2* knockout EBL cells following treatment with related-pathway activators, verified by plate counting. Data were relative to *RBMX2* knockout EBL cells untreated by activators.
(C, D) The impact of *M. bovis* BCG and *M. smegmatis* on the adhesion of *RBMX2* knockout EBL cells through plate counting assay. Data were relative to WT EBL cells after BCG and *M. Smegmatis* infection.
(E, F) The impact of *M. bovis* BCG and *M. smegmatis* on the invasion of *RBMX2* knockout EBL cells through plate counting assay. Data were relative to WT EBL cells after BCG and *M. Smegmatis* infection.
(G, H) The impact of *Salmonella* and *E. coli*, on the adhesion of *RBMX2* knockout EBL cells through plate counting assay. Data were relative to WT EBL cells after *Salmonella* and *E. coli* infection.
(K, L) The impact of *Salmonella* and *E. coli* on the invasion of *RBMX2* knockout EBL cells through plate counting assay. Data were relative to WT EBL cells after *Salmonella* and *E. coli* infection.
One-way and two-way ANOVA were used to determine the statistical significance of differences between different groups. Ns presents no significance, * presents $p < 0.05$, **presents $p < 0.01$, and *** presents $p < 0.001$ indicate statistically significant differences. Data were representative of at least three independent experiments.



Supplementary Fig 8

RBMX2 facilitates EMT process in EBL cells after *M. bovis*-infected BoMac cells.

(A) WB detection of changes in mesenchymal cell markers (*MMP-9* and *N-cadherin*) after infection of EBL cells with different infection ratios (10, 20, and 50) of *M. bovis*. Data were relative to EBL cells without infection of *M. bovis*.
(B) WB detection of the ability of *M. bovis* infection with macrophages (BoMac cells) to induce epithelial (EBL cells) mesenchymal transition. Data were relative to the addition of PBS in the upper chamber of the coculture model.
(C) Staining the skeleton of EBL cells in coculture model after *M. bovis* infection using ghost pen cyclic peptides. The cytoskeleton is labeled with red fluorescence, and the nucleus is stained with blue fluorescence. Scale Bar: 20 μm.
(D) EMT-related mRNAs (*MMP-9*, *N-cadherin*, and *E-cadherin*) expression was verified in coculture model EBL cells after *M. bovis* infection through RT-qPCR. Data were relative to coculture model EBL cells without *M. bovis* infection.
(E) The detection of EMT-related mRNAs (*MMP-9*, *N-cadherin*, and *E-cadherin*) of *RBMX2* knockout EBL cells after *M. bovis-infected* BoMac cells via RT-qPCR. Data were relative to WT EBL cells after *M. bovis-infected* BoMac cells.
Two-way ANOVA was used to determine the statistical significance of differences between different groups. Ns presents no significance, **presents $p < 0.01$, and *** presents $p < 0.001$ indicate statistically significant differences. Data were representative of at least three independent experiments.



Supplementary Fig 9

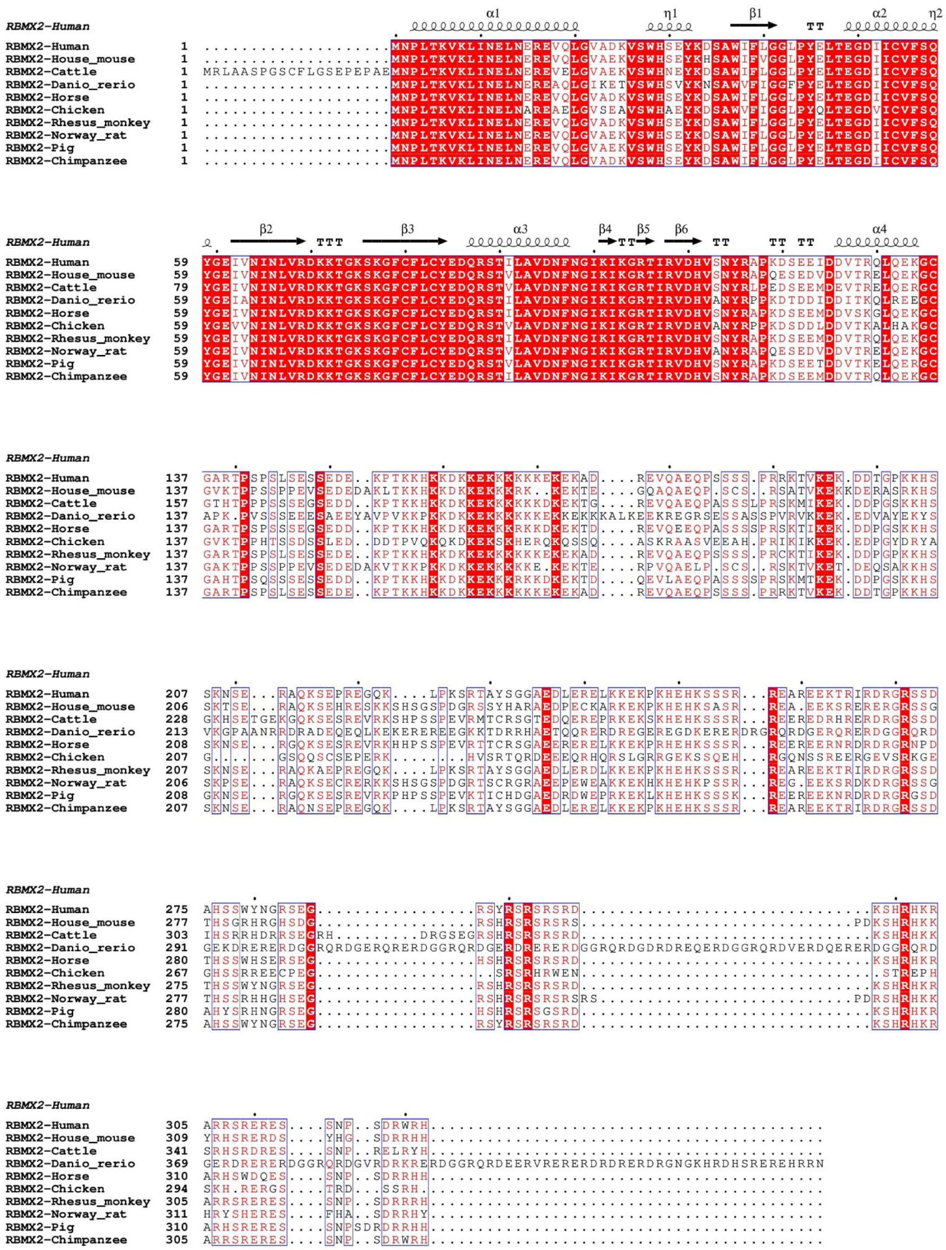
RBMX2 facilitates EMT in H1299 cells.

(A) EMT-related proteins expression was verified in H1299 cells after *M. bovis* infection through WB. Data were relative to H1299 Sh-Con cells.

(B) The change in the migratory and invasive capabilities of H1299 cells was assessed via Transwell assay. Data were relative to H1299 Sh-Con cells. Scale Bar: 100 μm.

(C) Activation of the MAPK pathway-related protein and p65 protein were activated after *RBMX2* knockout and WT EBL cells infected by *M. bovis* in this coculture model via WB. Data were relative to WT EBL cells with *M. bovis* infection.

Two-way ANOVA was used to determine the statistical significance of differences between different groups. *$p < 0.05$, **$p < 0.01$, and ***$p < 0.001$ indicate statistically significant differences. Data were representative of at least three independent experiments.



Supplementary Table 1

***RBMX2* exhibits high homology in amino acid sequence alignment across different species**

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Additional information

Author Contributions

YC and AG contributed to the conception and design of the study. CW, YP and HY carried out the experiment and wrote sections of the manuscript and made equal contribution. YJ, AK, ZY, KZ, and SX were involved in bacterial and cell research studies. CW, YP, HY, SX, LB, LZ, YY, and HC performed the statistical analysis. CW, AG, and YC wrote the manuscript with all authors providing feedback. All authors contributed to manuscript revision, proof-reading, and approval of the submitted version.

Abbreviations

- RBMX2: RNA-binding motif protein X-linked 2
- EBL: embryo bovine lung cells
- BoMac: bovine macrophage cells
- A549: human pulmonary alveolar epithelial cells
- TB: tuberculosis
- M. bovis: Mycobacterium bovis
- M. tb: Mycobacterium tuberculosis
- BCG: Bacillus Calmette–Guerin
- WOAH: World Organisation for Animal Health
- Hpi: hours postinfection
- M. smegmatis: Mycobacterium smegmatis
- E. coli: Escherichia coli
- p65 RELA: proto-oncogene
- NF-kB subunit: MAPK mitogen-activated protein kinase
- EM: Tepithelial–mesenchymal transition
- PARP: poly ADP-ribose polymerase
- ZO-1: tight junction protein 1
- CLDN-5: claudin 5
- OCLN: occludin
- MMP-9: matrix metallopeptidase 9
- SP-A: surfactant protein A
- GO: Gene Ontology
- KEGG: Kyoto Encyclopedia of Genes and Genomes
- WB: Western blot
- RT-qPCR: real-time quantitative polymerase chain reaction
- IF: immunofluorescence
- ECM: extracellular matrix

- TCGA: the cancer genome atlas

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Reviewer #1 (Public review):

The authors have addressed all my concerns and have addressed them to my satisfaction.

https://doi.org/10.7554/eLife.107132.2.sa2

Reviewer #3 (Public review):

Summary:

This study investigates the role of the host protein RBMX2 in regulating the response to Mycobacterium bovis infection and its connection to epithelial-mesenchymal transition (EMT), a key pathway in cancer progression. Using bovine and human cell models, the authors have wisely shown that RBMX2 expression is upregulated following M. bovis infection and promotes bacterial adhesion, invasion, and survival by disrupting epithelial tight junctions via the p65/MMP-9 signaling pathway. They also demonstrate that RBMX2 facilitates EMT and is overexpressed in human lung cancers, suggesting a potential link between chronic infection and tumor progression. The study highlights RBMX2 as a novel host factor that could serve as a therapeutic target for both TB pathogenesis and infection-related cancer risk.

Strengths:

The major strengths lie in its multi-omics integration (transcriptomics, proteomics, metabolomics) to map RBMX2's impact on host pathways, combined with rigorous functional assays (knockout/knockdown, adhesion/invasion, barrier tests) that establish causality through the p65/MMP-9 axis. Validation across bovine and human cell models and in clinical tissue samples enhances translational relevance. Finally, identifying RBMX2 as a novel regulator linking mycobacterial infection to EMT and cancer progression opens exciting therapeutic avenues.

Weaknesses:

There are a few minor weaknesses like grammatical errors, spelling mistakes. Also, the manuscript is too dense; improving the narratives in the Results and Discussion section could help readers follow the logic of the experimental design and conclusions.

https://doi.org/10.7554/eLife.107132.2.sa1

Author response:

The following is the authors’ response to the original reviews.

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *This manuscript presents a compelling study identifying RBMX2 as a novel host factor upregulated during Mycobacterium bovis infection.*
>
> *The study demonstrates that RBMX2 plays a role in:*

> *(1) Facilitating M. bovis adhesion, invasion, and survival in epithelial cells.*
>
> *(2) Disrupting tight junctions and promoting EMT.*
>
> *(3) Contributing to inflammatory responses and possibly predisposing infected tissue to lung cancer development.*
>
> *By using a combination of CRISPR-Cas9 library screening, multi-omics, coculture models, and bioinformatics, the authors establish a detailed mechanistic link between M. bovis infection and cancer-related EMT through the p65/MMP-9 signaling axis. Identification of RBMX2 as a bridge between TB infection and EMT is novel.*
>
> *Strengths:*
>
> *This topic and data are both novel and significant, expanding the understanding of transcriptomic diversity beyond RBM2 in M. bovis responsive functions.*
>
> *Weaknesses:*
>
> *(1) The abstract and introduction sometimes suggest RBMX2 has protective anti-TB functions, yet results show it facilitates pathogen adhesion and survival. The authors need to rephrase claims to avoid contradiction.*

We sincerely appreciate the reviewer's valuable feedback regarding the need to clarify RBMX2's role throughout the manuscript. We have carefully revised the text to ensure consistent messaging about RBMX2's function in promoting M. bovis infection. Below we detail the specific modifications made:\

(1) Introduction Revisions:

Changed "The objective of this study was to elucidate the correlation between host genes and the susceptibility of M.bovis infection" to "The objective of this study was to identify host factors that promote susceptibility to M.bovis infection"

Revised "RBMX2 polyclonal and monoclonal cell lines exhibited favorable phenotypes" to "RBMX2 knockout cell lines showed reduced bacterial survival"

Replaced "The immune regulatory mechanism of RBMX2" with "The role of RBMX2 in facilitating M.bovis immune evasion"

(2) Results Revisions:

Modified "RBMX2 fails to affect cell morphology and the ability to proliferate and promotes M.bovis infection" to "RBMX2 does not alter cell viability but significantly enhances M.bovis infection"

Strengthened conclusion in Figure 4: "RBMX2 actively disrupts tight junctions to facilitate bacterial invasion"

(3) Discussion Revisions:

Revised screening description: "We screened host factors affecting M.bovis susceptibility and identified RBMX2 as a key promoter of infection"

Strengthened concluding statement: "In summary, RBMX2 drives TB pathogenesis by compromising epithelial barriers and inducing EMT"

These targeted revisions ensure that:

All sections consistently present RBMX2 as promoting infection; the language aligns with our experimental finding; potential protective interpretations have been eliminated. We believe these modifications have successfully addressed the reviewer's concern while maintaining the manuscript's original structure and scientific content. We appreciate the opportunity to improve our manuscript and thank the reviewer for this constructive suggestion.

> *(2) While p65/MMP-9 is convincingly implicated, the role of MAPK/p38 and JNK is less clearly resolved.*

We sincerely appreciate the reviewer's insightful comment regarding the roles of MAPK/p38 and JNK in our study. Our experimental data clearly demonstrated that RBMX2 knockout significantly reduced phosphorylation levels of p65, p38, and JNK (Fig. 5A), indicating potential involvement of all three pathways in RBMX2-mediated regulation.

Through systematic functional validation, we obtained several important findings:

In pathway inhibition experiments, p65 activation (PMA treatment) showed the most dramatic effects on both tight junction disruption (ZO-1, OCLN reduction) and EMT marker regulation (E-cadherin downregulation, N-cadherin upregulation);p38 activation (ML141 treatment) exhibited moderate effects on these processes; JNK activation (Anisomycin treatment) displayed minimal impact.

Most conclusively, siRNA-mediated silencing of p65 alone was sufficient to:

Restore epithelial barrier function

Reverse EMT marker expression

Reduce bacterial adhesion and invasion

These results establish a clear hierarchy in pathway importance: p65 serves as the primary mediator of RBMX2's effects, while p38 plays a secondary role and JNK appears non-essential under our experimental conditions. We have now clarified this relationship in the revised Discussion section to strengthen this conclusion.

This refined understanding of pathway hierarchy provides important mechanistic insights while maintaining consistency with all our experimental data. We thank the reviewer for this valuable suggestion that helped improve our manuscript.

> *(3) Metabolomics results are interesting but not integrated deeply into the main EMT narrative.*

Thank you for this constructive suggestion. In this article, we detected the metabolome of RBMX2 knockout and wild-type cells after Mycobacterium bovis infection, which mainly served as supporting evidence for our EMT model. However, we did not conduct an in-depth discussion of these findings. We have now added a detailed discussion of this section to further support our EMT model.

ADD:Meanwhile, metabolic pathways enriched after RBMX2 deletion, such as nucleotide metabolism, nucleotide sugar synthesis, and pentose interconversion, primarily support cell proliferation and migration during EMT by providing energy precursors, regulating glycosylation modifications, and maintaining redox balance; cofactor synthesis and amino sugar metabolism participate in EMT regulation through influencing metabolic remodeling and extracellular matrix interactions; chemokine and cGMP-PKG signaling pathways may further mediate inflammatory responses and cytoskeletal rearrangements, collectively promoting the EMT process.

(4) A key finding and starting point of this study is the upregulation of RBMX2 upon M. bovis infection. However, the authors have only assessed RBMX2 expression at the mRNA level following infection with M. bovis and BCG. To strengthen this conclusion, it is essential to validate RBMX2 expression at the protein level through techniques such as Western blotting or immunofluorescence. This would significantly enhance the credibility and impact of the study's foundational observation.

Thank you for your comment. We have supplemented the experiments in this part and found that Mycobacterium bovis infection can significantly enhance the expression level of RBMX2 protein.

(5) The manuscript would benefit from a more in-depth discussion of the relationship between tuberculosis (TB) and lung cancer. While the study provides experimental evidence suggesting a link via EMT induction, integrating current literature on the epidemiological and mechanistic connections between chronic TB infection and lung tumorigenesis would provide important context and reinforce the translational relevance of the findings.

We sincerely appreciate the valuable comments from the reviewer. We fully agree with your suggestion to further explore the relationship between tuberculosis (TB) and lung cancer. In the revised manuscript, we will add a new paragraph in the Discussion section to systematically integrate the current literature on the epidemiological and mechanistic links between chronic tuberculosis infection and lung cancer development, including the potential bridging roles of chronic inflammation, tissue damage repair, immune microenvironment remodeling, and the epithelial-mesenchymal transition (EMT) pathway. This addition will help more comprehensively interpret the clinical implications of the observed EMT activation in the context of our study, thereby enhancing the biological plausibility and clinical translational value of our findings.

ADD:There is growing epidemiological evidence suggesting that chronic TB infection represents a potential risk factor for the development of lung cancer. Studies have shown that individuals with a history of TB exhibit a significantly increased risk of lung cancer, particularly in areas of the lung with pre-existing fibrotic scars, indicating that chronic inflammation, tissue repair, and immune microenvironment remodeling may collectively contribute to malignant transformation 74. Moreover, EMT not only endows epithelial cells with mesenchymal features that enhance migratory and invasive capacity but is also associated with the acquisition of cancer stem cell-like properties and therapeutic resistance 75. Therefore, EMT may serve as a crucial molecular link connecting chronic TB infection with the malignant transformation of lung epithelial cells, warranting further investigation in the intersection of infection and tumorigenesis.

Reviewer #2 (Public review):

Summary:

I am not familiar with cancer biology, so my review mainly focuses on the infection part of the manuscript. Wang et al identified an RNA-binding protein RBMX2 that links the Mycobacterium bovis infection to the epithelial-Mesenchymal transition and lung cancer progression. Upon mycobacterium infection, the expression of RBMX2 was moderately increased in multiple bovine and human cell lines, as well as bovine lung and liver tissues. Using global approaches, including RNA-seq and proteomics, the authors identified differential gene expression caused by the RBMX2 knockout during M. bovis infection. Knockout of RBMX2 led to significant upregulations of tight-junction related genes such as CLDN-5, OCLN, ZO-1, whereas M. bovis infection affects the integrity of

> *epithelial cell tight junctions and inflammatory responses. This study establishes that RBMX2 is an important host factor that modulates the infection process of M. bovis.*
>
> *Strengths:*
>
> *(1) This study tested multiple types of bovine and human cells, including macrophages, epithelial cells, and clinical tissues at multiple timepoints, and firmly confirmed the induced expression of RBMX2 upon M. bovis infection.*
>
> *(2) The authors have generated the monoclonal RBMX2 knockout cell lines and comprehensively characterized the RBMX2-dependent gene expression changes using a combination of global omics approaches. The study has validated the impact of RBMX2 knockout on the tight-junction pathway and on the M. bovis infection, establishing RBMX2 as a crucial host factor.*
>
> *Weaknesses:*
>
> *(1) The RBMX2 was only moderately induced (less than 2-fold) upon M. bovis infection, arguing its contribution may be small. Its value as a therapeutic target is not justified. How RBMX2 was activated by M. bovis infection was unclear.*

Thank you for your valuable and constructive comments. In this study, we primarily utilized the CRISPR whole-genome screening approach to identify key factors involved in bovine tuberculosis infection. Through four rounds of screening using a whole-genome knockout cell line of bovine lung epithelial cells infected with Mycobacterium bovis, we identified RBMX2 as a critical factor.

Although the transcriptional level change of RBMX2 was less than two-fold, following the suggestion of Reviewer 1, we examined its expression at the protein level, where the change was more pronounced, and we have added these results to the manuscript.

Regarding the mechanism by which RBMX2 is activated upon M. bovis infection, we previously screened for interacting proteins using a Mycobacterium tuberculosis secreted and membrane protein library, but unfortunately, we did not identify any direct interacting proteins from M. tuberculosis (https://doi.org/10.1093/nar/gkx1173).

> *(2) Although multiple time points have been included in the study, most analyses lack temporal resolution. It is difficult to appreciate the impact/consequence of M. bovis infection on the analyzed pathways and processes.*

We appreciate the valuable comments from the reviewers. Although our study included multiple time points post-infection, in our experimental design we focused on different biological processes and phenotypes at distinct time points:

During the early phase (e.g., 2 hours post-infection), we focused on barrier phenotypes during the intermediate phase (e.g., 24 hours post-infection), we concentrated more on pathway activation and EMT phenotypes;

And during the later phase (e.g., 48–72 hours post-infection), we focused more on cell death phenotypes, which were validated in another FII article (https://doi.org/10.3389/fimmu.2024.1431207).

We also examined the impact of varying infection durations on RBMX2 knockout EBL cellular lines via GO analysis. At 0 hpi, genes were primarily related to the pathways of cell junctions, extracellular regions, and cell junction organization. At 24 hpi, genes were mainly associated with pathways of the basement membrane, cell adhesion, integrin binding and cell migration By 48 hpi, genes were annotated into epithelial cell differentiation and were negatively regulated during epithelial cell proliferation. This indicated that RBMX2 can regulate cellular connectivity throughout the stages of M. bovis infection.

For KEGG analysis, genes linked to the MAPK signaling pathway, chemical carcinogen-DNA adducts, and chemical carcinogen-receptor activation were observed at 0 hpi. At 24 hpi, significant enrichment was found in the ECM-receptor interaction, PI3K-Akt signaling pathway, and focal adhesion. Upon enrichment analysis at 48 hpi, significant enrichment was noted in the TGF-beta signaling pathway, transcriptional misregulation in cancer, microRNAs in cancer, small cell lung cancer, and p53 signaling pathway.

> ***Reviewer #3 (Public review):***
>
> *Summary:*
>
> *This study investigates the role of the host protein RBMX2 in regulating the response to Mycobacterium bovis infection and its connection to epithelial-mesenchymal transition (EMT), a key pathway in cancer progression. Using bovine and human cell models, the authors have wisely shown that RBMX2 expression is upregulated following M. bovis infection and promotes bacterial adhesion, invasion, and survival by disrupting epithelial tight junctions via the p65/MMP-9 signaling pathway. They also demonstrate that RBMX2 facilitates EMT and is overexpressed in human lung cancers, suggesting a potential link between chronic infection and tumor progression. The study highlights RBMX2 as a novel host factor that could serve as a therapeutic target for both TB pathogenesis and infection-related cancer risk.*
>
> *Strengths:*
>
> *The major strengths lie in its multi-omics integration (transcriptomics, proteomics, metabolomics) to map RBMX2's impact on host pathways, combined with rigorous functional assays (knockout/knockdown, adhesion/invasion, barrier tests) that establish causality through the p65/MMP-9 axis. Validation across bovine and human cell models and in clinical tissue samples enhances translational relevance. Finally, identifying RBMX2 as a novel regulator linking mycobacterial infection to EMT and cancer progression opens exciting therapeutic avenues.*
>
> *Weaknesses:*
>
> *Although it's a solid study, there are a few weaknesses noted below.*
>
> *(1) In the transcriptomics analysis, the authors performed (GO/KEGG) to explore biological functions. Did they perform the search locally or globally? If the search was performed with a global reference, then I would recommend doing a local search. That would give more relevant results. What is the logic behind highlighting some of the enriched pathways (in red), and how are they relevant to the current study?*

We appreciate the reviewer's thoughtful questions regarding our transcriptomic analysis. In this study, we employed a localized enrichment approach focusing specifically on gene expression profiles from our bovine lung epithelial cell system. This cell-type-specific analysis provides more biologically relevant results than global database searches alone.

Regarding the highlighted pathways, these represent:

Temporally significant pathways showing strongest enrichment at each stage:

(1) 0h: Cell junction organization (immediate barrier response)

(2) 24h: ECM-receptor interaction (early EMT initiation)

(3) 48h: TGF-β signaling (chronic remodeling)

Mechanistically linked to our core findings about RBMX2's role in:

(1) Epithelial barrier disruption

(2) Mesenchymal transition

(3) Chronic infection outcomes

We selected these particular pathways because they:

(1) Showed the most statistically significant changes (FDR <0.001)

(2) Formed a coherent biological narrative across infection stages

(3) Were independently validated in our functional assays

This targeted approach allows us to focus on the most infection-relevant pathways while maintaining statistical rigor.

> *(2) While the authors show that RBMX2 expression correlates with EMT-related gene expression and barrier dysfunction, the evidence for direct association remains limited in this study. How does RBMX2 activate p65? Does it bind directly to p65 or modulate any upstream kinases? Could ChIP-seq or CLIP-seq provide further evidence for direct RNA or DNA targets of RBMX2 that drive EMT or NF-κB signaling?*

We sincerely appreciate the reviewer's in-depth questions regarding the mechanisms by which RBMX2 activates p65 and its association with EMT. Although the molecular mechanism remains to be fully elucidated, our study has provided experimental evidence supporting a direct regulatory relationship between RBMX2 and the p65 subunit of the NF-κB pathway. Specifically, we investigated whether the transcription factor p65 could directly bind to the promoter region of RBMX2 using CHIP experiments. The results demonstrated that the transcription factor p65 can physically bind to the RBMX2 region.

Furthermore, dual-luciferase reporter assays were conducted, showing that p65 significantly enhances the transcriptional activity of the RBMX2 promoter, indicating a direct regulatory effect of RBMX2 on p65 expression.

These findings support our hypothesis that RBMX2 activates the NF-κB signaling pathway through direct interaction with the p65 protein, thereby participating in the regulation of EMT progression and barrier function.

In our subsequent work papers, we will also employ experiments such as CLIP to further investigate the specific mechanisms through which RBMX2 exerts its regulatory functions.

ADD and Revise in Results:

To thoroughly verify the regulatory mechanism between RBMX2 and p65, we initiated our investigation by conducting an in-depth analysis of the RBMX2 promoter region to identify potential interactions with the transcription factor p65. Initially, we performed molecular docking simulations to predict the binding affinity and interaction patterns between RBMX2 and p65 proteins. These simulations revealed multiple amino acid residues within the RBMX2 protein that formed strong, stable interactions with p65. The docking analysis yielded a high docking score of 1978.643 (Fig. 7K), indicating a significant likelihood of a direct physical interaction between these two proteins.

To complement the protein-protein interaction analysis, we next investigated whether p65 could directly bind to the promoter region of the RBMX2 gene at the transcriptional level. Using the JASPAR database, a comprehensive resource for transcription factor binding profiles, we queried the RBMX2 promoter sequence for potential p65 binding sites. This analysis identified several putative binding motifs, suggesting that p65 may act as a transcriptional regulator of RBMX2 expression.

To experimentally validate this transcriptional regulatory relationship, we employed a dual-luciferase reporter assay. We cloned the RBMX2 promoter region containing the predicted p65 binding sites into a luciferase reporter plasmid. This construct was then co-transfected into cultured cells along with a plasmid expressing p65. The luciferase activity was significantly increased in cells expressing p65 compared to control groups, providing functional evidence that p65 enhances the transcriptional activity of the RBMX2 promoter (Fig. 7I).

Furthermore, to confirm the direct binding of p65 to the RBMX2 promoter in a chromatin context, we performed chromatin immunoprecipitation followed by quantitative PCR (ChIP-qPCR). In this assay, we used specific antibodies against p65 to immunoprecipitate chromatin fragments containing p65-bound DNA. The enriched DNA fragments were then analyzed using primers targeting the RBMX2 promoter region. Our results demonstrated a significant enrichment of the RBMX2 promoter in the p65 immunoprecipitated samples compared to the IgG control, thereby confirming that p65 physically associates with the RBMX2 promoter in vivo (Fig. 7J). Collectively, these findings-ranging from computational docking predictions to transcriptional reporter assays and ChIP validation-provide strong evidence supporting a direct regulatory interaction between p65 and RBMX2. This regulatory mechanism may play a critical role in the biological pathways involving these two molecules, particularly in contexts such as inflammation, immune response, or cellular stress, where p65 (a subunit of NF-κB) is known to be prominently involved.

> *(3) The manuscript suggests that RBMX2 enhances adhesion/invasion of several bacterial species (e.g., E. coli, Salmonella), not just M. bovis. This raises questions about the specificity of RBMX2's role in Mycobacterium-specific pathogenesis. Is RBMX2 a general epithelial barrier regulator or does it exhibit preferential effects in mycobacterial infection contexts? How does this generality affect its potential as a TB-specific therapeutic target?*

Thank you for your valuable comments. When we initially designed this experiment, we were interested in whether the RBMX2 knockout cell line could confer effective resistance not only against Mycobacterium bovis but also against Gram-negative and Gram-positive bacteria. Surprisingly, we indeed observed resistance to the invasion of these pathogens, albeit weaker compared to that against Mycobacterium bovis.

Nevertheless, we believe these findings merit publication in eLife. Moreover, RBMX2 knockout does not affect the phenotype of epithelial barrier disruption under normal conditions; its significant regulatory effect on barrier function is only evident upon infection with Mycobacterium bovis.

Importantly, during our genome-wide knockout library screening, RBMX2 was not identified in the screening models for Salmonella or Escherichia coli, but was consistently detected across multiple rounds of screening in the Mycobacterium bovis model.

> *(4) The quality of the figures is very poor. High-resolution images should be provided.*

Thank you for your feedback; we provided higher-resolution images.

> *(5) The methods are not very descriptive, particularly the omics section.*

Thank you for your comments; we have revised the description of the sequencing section.

> *(6) The manuscript is too dense, with extensive multi-omics data (transcriptomics, proteomics, metabolomics) but relatively little mechanistic integration. The authors should have focused on the key mechanistic pathways in the figures. Improving the narratives in the Results and Discussion section could help readers follow the logic of the experimental design and conclusions.*

Thank you for your valuable comments. We have streamlined the figures and revised the description of the results section accordingly.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *(1) The first part of the results and the major conclusions largely overlap with the previous paper by the same authors (Frontiers in Immunology, https://doi.org/10.3389/fimmu.2024.1431207). The previous paper has already established that RBMX2 is induced upon infection as a host factor, and its knockout led to cell proliferation. Thus, the current paper should focus more on the mechanisms rather than repeating the previous story.*

We appreciate the reviewer's careful reading and constructive feedback. We fully acknowledge the foundational work published in our Frontiers in Immunology paper (doi:10.3389/fimmu.2024.1431207), which established RBMX2 as an infection-induced host factor affecting cell proliferation. The current study represents a significant mechanistic extension of these initial findings, with the following key advances:

(1) Novel Mechanistic Insights (Current Study Focus):

Discovery of the p65/MMP-9 pathway as the central mechanism mediating RBMX2's effects on EMT (Figs. 4-6)

First demonstration of RBMX2's role in epithelial barrier disruption (Figs. 2-3)

Identification of temporal regulation patterns during infection progression (Fig. 7)

(2) Expanded Biological Scope:

Demonstration of RBMX2's function in both bovine and human cell systems (vs. previous bovine-only data)

Clinical correlation with TB lesions

Therapeutic potential assessment through pathway inhibition

(3) Technical Advancements:

CRISPR-based mechanistic validation (vs. previous siRNA approach)

Multi-omics integration (transcriptomics + metabolomics)

Advanced live-cell imaging

We have now:

Removed redundant proliferation data from Results

Sharpened the Introduction to highlight mechanistic questions

Added explicit discussion comparing both studies

The current work provides the first comprehensive mechanistic framework for RBMX2's role in TB pathogenesis, moving substantially beyond the initial observational findings. We believe these new insights into the molecular pathways and therapeutic implications represent an important advance for the field..

> *(2) Line 107-110: The CRISPR screening results are not provided. Has it been published, or is it an unpublished dataset? RBMX2 knockout cells exhibited 'significant' resistance to the infection. How significant? Data?*

Thank you for your valuable comments. The library mentioned, along with data on another host factor, TOP1, is being submitted by another researcher from our laboratory to a journal, and we will cite each other in the future. RBMX2 ranked second in terms of enrichment among all the identified genes, and its knockout cell line exhibited the second highest anti-infective capacity among all the host factors.

> *(3) Line 152: The RNA-seq analysis has already been performed/reported in the previous Frontiers paper. Therein, 173 genes were found to be differentially expressed. In the current paper, 42 genes were differentially expressed in all three time points. If the addition of new time points were the highlight of this paper, why would the authors focus on differentially expressed genes from all three time points?*

Thank you for your valuable comments.

In the newly added data, we aimed to investigate the temporal changes during Mycobacterium bovis infection of host cells.

Previous study (Frontiers): Single 24h timepoint → 173 DEGs

Current study: Three timepoints (0h, 24h, 48h) with 42 consistently regulated genes → Reveals temporally stable core regulators of infection response

On one hand, we briefly described in the manuscript those important genes that exhibited changes across all time points.

On the other hand, in the supplementary materials, we also focused on the enriched genes at each individual time point, to better understand the temporal dynamics regulated by RBMX2.

> *(4) Line 153: The '0 h' time point is in fact 2 h post-infection. Why did the authors skip the real 0h time point? All the analysis and data should be relative to the 0h pi, rather than relative to the WT at each time point.*

We appreciate the reviewer's important question regarding our timepoint nomenclature. The experimental timeline was designed as follows:

(1) Infection Protocol:

2h to 0h: Bacterial co-culture (MOI 20:1)

0h: Gentamicin (100 μg/ml) added to kill extracellular bacteria

0h+: Monitored intracellular survival

(2) Rationale for "0h" Designation:

This marks the onset of intracellular infection phase when Extracellular bacteria are eliminated (validated by plating)Host cell responses to intracellular pathogens begin All subsequent measurements reflect genuine infection (not attachment)

(3)Technical Validation:

Confirmed complete extracellular killing by:

Culture supernatant plating (0 CFU after gentamycin)

Microscopy (no surface-associated bacteria)

(4) Comparative Analysis:

All data are presented as:

Fold-change relative to uninfected controls at each timepoint

We have now:

Clarified the timeline in Methods

Specified "0h = post-gentamicin" in all figure legends

This standardized approach aligns with established intracellular pathogen studies (e.g., Cell Microbiol. 2018;20:e12840). We're happy to adjust terminology if "0hpi (post-invasion)" would be clearer.

> *(5) Figure 2F: The data should be compared to the 0h pi, and show the temporal changes of gene expression.*

Thank you for your suggestion. We have added additional information to this section. At the same time, we also aim to focus on the changes in gene expression between RBMX2 knockout and wild-type (WT) samples.

We have now:

Added temporal expression profiles relative to 0hpi baseline (SFig.4C).

Clarified the dual normalization approach in Methods

Maintained original between-group comparisons for phenotypic correlation

> *(6) Line 207. Not all the proteins were down-regulated post-infection.*

Thank you for your comment. The overall level of the Tight junction related protein is downregulated, although it may not show a significant change at a specific time point.

We have revised our description, changing the keyword from "All" to "Most."

> *(7) Line 278, the introduction of the H1299 cell line should appear earlier when it was mentioned for the first time in the manuscript.*

Thank you for your comment. We have provided a description in the abstract and Result1.

ADD:

Abstrat: Meanwhile, we also validated the EMT process in human lung epithelial cancer cells H1299.

Result 1: Furthermore, RBMX2-silenced H1299 cells exhibited a higher survival rate compared to H1299 ShNc cells after M. bovis infection (Fig. 1H).

> *(8) Figure 4 is huge and almost illegible, which may be divided into two figures.*

Thank you for your valuable comments. We have streamlined the figures and revised the description of the results section accordingly.

> ***Reviewer #3 (Recommendations for the authors):***
>
> *I encountered frequent grammatical and syntactic issues. Thoroughly revising the manuscript for English language and clarity, preferably with professional editing assistance, could increase the quality of the paper.*

Thank you for your valuable comments; we will invite a professional editor to polish the language.

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